Prototype-Guided Multilayer Alignment Network for Few-Shot Object Detection in Remote Sensing

Abdullah Azeem, Zhengzhou Li, *Member, IEEE*, Abubakar Siddique, Yuting Zhang, and Yongsong Li

***Abstract*— Few-shot object detection (FSOD) addresses the challenge of limited labeled data by enabling detectors to learn from minimal annotations. Recent work on image–text fusion has shown promise in overcoming data scarcity issues. However, these models suffer from catastrophic forgetting due to two distinct challenges during few-shot adaptation. First, feature space distortion disrupts established relationships between image–text modalities and blurs decision boundaries, leading to inaccuracies in object classification. Second, attention drift impairs the precise visual–textual alignments learned during base training, limiting the model's ability to focus on relevant regions for accurate object localization. To overcome these limitations, we propose Protoalign, a novel prototype-guided multilayer alignment network that maintains robust cross-modal relationships across multiple network layers while mitigating catastrophic forgetting. Specifically, cross-modal prototype guidance (CPG) enables stable feature learning through class-specific prototype fusion to mitigate feature space distortion. Multimodal feature aggregation (MFA) strengthens feature relationships through channel-level interactions to overcome attention drift. Moreover, we propose an integrator that facilitates consistent information flow between network layers. Protoalign progressively refines multimodal features, enabling effective novel class adaptation while preserving crucial base knowledge. The refined representations are then processed by a detection transformer (DETR) decoder for accurate object detection. Extensive experiments on iSAID, DIOR, FAIR1M-Airplane, and NWPU VHR-10 datasets demonstrate that Protoalign achieves superior performance while significantly reducing catastrophic forgetting compared to existing methods.**

***Index Terms*— Few-shot learning, multimodal fusion, object detection, remote sensing, transformer.**

I. Introduction

OBJECT detection systems often encounter difficulties when learning from limited data, a challenge prevalent in many real-world applications from anomaly detection [1], [2] to environmental monitoring [3], [4]. Few-shot object detection (FSOD) has emerged as a promising solution, enabling models to detect novel objects using only a few examples. However, existing methods face significant hurdles during few-shot learning. For example, meta-learning approaches [5], [6], [7] aim to develop adaptation strategies based on base classes but often struggle to transfer this knowledge effectively. On the other hand, fine-tuning strategies [8], [9], [10], [11] attempt to adapt pretrained models to new classes but typically fail to maintain stable representations.

Remote sensing images present unique challenges for FSOD that significantly amplify the difficulty compared to natural images. Unlike natural images captured horizontally from ground level with uniform object orientations and centralized compositions, remote sensing images are acquired from a bird's-eye view that alters object characteristics. This perspective difference introduces arbitrary object orientations regardless of category [Fig. 1(a)], extreme scale variations dependent on captured altitude [Fig. 1(b)], sparse spatial distribution patterns [Fig. 1(d)], and environmental interference factors, including cloud-induced occlusion [Fig. 1(c)]. These characteristics create a challenging environment where objects from different classes exhibit minimal interclass differences while simultaneously displaying high intraclass diversity. When combined with the data scarcity in remote sensing, these factors severely limit the effectiveness of conventional few-shot detection approaches that rely solely on visual features, as models struggle to develop stable and discriminative representations that can generalize effectively across the unique challenges presented by remote sensing imagery.

To address remote sensing FSOD challenges, researchers have developed specialized approaches that fall into two main categories. Meta-learning methods develop adaptation mechanisms that efficiently transfer knowledge with minimal examples. These approaches [12], [13], [14], [15] focus on learning to extract class-agnostic features during base training that can quickly adapt to novel categories. Fine-tuning approaches adapt pretrained models to enhance the performance on novel classes with limited examples. These methods target specific challenges such as incomplete annotations [16], background–foreground confusion [17], and high feature variance across similar objects [18], often employing selective freezing strategies to maintain critical visual knowledge while reducing computational costs. Despite these advancements, both approaches still struggle to maintain stable representations when learning from limited examples of novel objects, largely due to their exclusive reliance on visual features.

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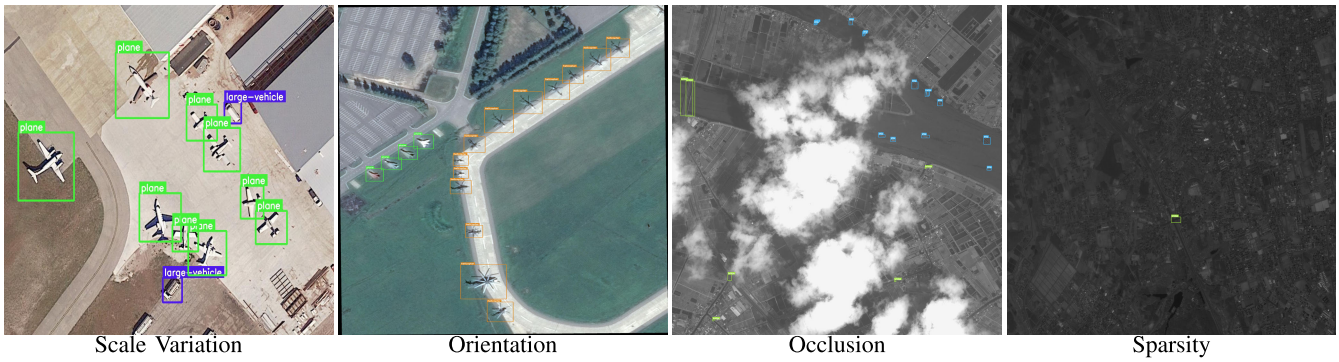


Fig. 1. Remote sensing images typically exhibit the following characteristics: object orientations, scale variability, occlusion, and sparsity.

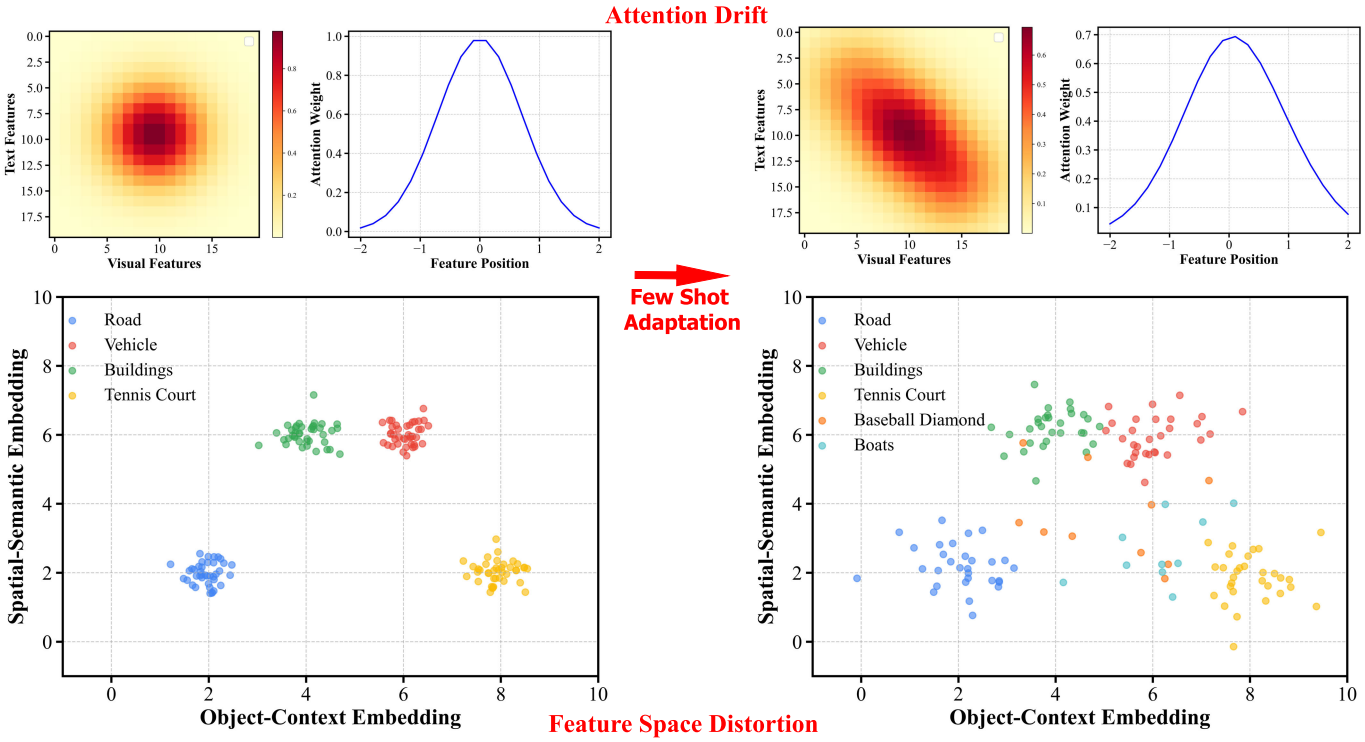


Fig. 2. Two degradation patterns during adaptation from base classes (road, vehicle, buildings, and tennis court) to novel classes (baseball diamond and boats). (Left) During base training, the model learns stable cross-modal alignments between visual and textual features. (Right) During few-shot fine-tuning, these alignments deteriorate through two mechanisms: 1) feature space distortion, shown through scattered feature embeddings in the t-SNE plots, where class clusters become less distinct after adaptation and 2) attention drift, visualized through attention heatmaps that lose focus on discriminative object regions.

Incorporating text with images enhances models' ability to detect small, obscure, or ambiguous objects by providing rich contextual information from textual [19]. Recent research has explored various image–text fusion methods to tackle data scarcity in FSOD [20], [21], [22]. Additionally, multilayer fusion architectures [23], [24], [25] have shown promise in strengthening multimodal alignments by integrating features across multiple layers. However, despite the advantages of image–text fusion, significant catastrophic forgetting occurs during the adaptation to new classes in FSOD. Although some incremental learning approaches [26], [27], [28], [29] have successfully retained features of base classes, their effectiveness in adapting to new categories is often limited. This limitation arises from an excessive focus on preserving base class features, coupled with a lack of rich semantic information from text when working with limited data.

The challenges of applying image–text fusion in FSOD largely stem from catastrophic forgetting, which manifests as feature space distortion and attention drift. Feature space distortion results in the degradation of the model's internal representations, corrupting the discriminative representations. During base training, object categories are mapped to well-separated feature regions. However, adapting to novel classes destabilizes these relationships, causing feature clusters to shift and decision boundaries to become blurred. Second, attention drift disrupts the precise alignments between visual and textual features learned during base training. As shown in Fig. 2, base training establishes precise cross-modal attention patterns. However, novel classes cause attention patterns to

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drift, making it difficult for the model to maintain focus on relevant image regions for both base and novel class descriptions. Consequently, the model struggles to differentiate what it is looking at (due to distorted features) and where it should be looking (due to attention drift), making accurate object detection significantly more difficult. While feature space distortion impacts the model's core representational ability, attention drift compromises how the model contextualizes those representations using associated text, highlighting the need to stabilize both during adaptation.

To tackle these challenges, we introduce the prototype-guided multilayer alignment network (Protoalign), an innovative architecture designed to mitigate catastrophic forgetting through prototype-guided hierarchical feature alignment between visual and textual modalities. During the base training phase, Protoalign systematically refines and aligns representations across the network's layers, resulting in robust features that are less susceptible to degradation during adaptation. Specifically, Protoalign employs cross-modal prototype guidance (CPG) to process image and text embeddings, establishing class-specific feature anchors that help counteract feature space distortion. By creating stable reference points through prototypes, CPG effectively constrains feature drift during adaptation, preserving the decision boundaries of base classes while accommodating new categories. These prototypes aggregate discriminative patterns across instances, forming class representations that are resilient to individual variations. To address attention drift, we implement multimodal feature aggregation (MFA), which enhances semantic relationships through channel-level interactions. Instead of relying on token-level attention, MFA examines channel correlations to identify stable visual–textual relationships, effectively bridging the semantic gap between modalities and facilitating effective reasoning across them. The integrator module ensures a consistent flow of information between layers, promoting smooth information propagation throughout the network. These cross-modal representations are then processed by a detection transformer (DETR) decoder for accurate object detection. Protoalign demonstrates superior performance compared to existing methods on iSAID [30], DIOR [31], FAIR1M-Airplane [32], and NWPU VHR-10 [3] datasets.

In this article, we make the following contributions.

1) Protoalign tackles catastrophic forgetting caused by feature space distortion and attention drift by ensuring stable cross-modal representations across multiple network layers during few-shot adaptation.
2) The CPG module introduces a novel mechanism for stable feature learning using class-specific prototypes to address feature space distortion. It uses a gated asymmetric attention design, where visual features actively query textual information and separate refinement pathways to preserve modality-specific characteristics.
3) MFA captures consistent cross-modal patterns by analyzing channel-level correlations to address attention drift. MFA identifies stable relationships between visual attributes and textual descriptions, improving cross-modal alignment and bridging the semantic gap through channel-level interactions.

II. Related Work

Object Detection: It has progressed through various architectural paradigms. Early approaches relied on handcrafted features and filtering mechanisms [33], yet frequently encountered difficulties in complex environments due to parameter sensitivity limitations. The advent of deep learning has substantially transformed object detection performance. Faster R-CNN [34] enhanced region proposal mechanisms and multiscale feature representation, albeit at considerable computational cost. Building on this foundation, feature pyramid network (FPN) [35] incorporated feature pyramid architectures to enable multiscale object detection, though it maintains comparable computational demands. The you only look once (YOLO) framework [36] tackled efficiency concerns through unified single-stage detection but exhibited limitations with orientation-variant objects. CenterNet [37] enhanced detection precision by adopting keypoint-based object localization strategies. The introduction of DETR [38] transformed object detection by treating it as a direct set prediction problem using a transformer encoder–decoder architecture, eliminating hand-designed components like anchor boxes and nonmaximum suppression through bipartite matching, while its enhanced version Deformable DETR [39] incorporated deformable attention modules focusing on key sampling points for improved computational efficiency and small object detection. DINO [40] incorporated contrastive denoising training and mixed query selection mechanisms, Conditional-DETR [41] enhanced query mechanisms for improved training efficiency, and DAB-DETR [42] introduced dynamic anchor boxes as queries. To specifically address orientation challenges in remote sensing and aerial imagery, recent transformer-based approaches have emerged, including OrientedFormer [43], which presents an end-to-end transformer architecture specifically designed for oriented object detection in remote sensing imagery; ARS-DETR [44], which introduces aspect ratio-sensitive detection mechanisms within the transformer framework for oriented object detection; and EMO2-DETR [45], which focuses on efficient matching strategies for oriented objects using transformer architectures.

Despite these advancements, these approaches face a shared challenge in requiring large volumes of annotated data, creating significant obstacles for domain-specific implementations.

A. FSOD in Remote Sensing

In contrast to conventional object detection, which requires a large volume of annotated data, FSOD enables models to detect and classify novel object categories using only a few annotated examples, typically leveraging prior knowledge learned from well-annotated base classes. Recent research on FSOD has explored various approaches to tackle the challenge of learning from limited data, with distinct strategies emerging for both natural images and remote sensing. In natural images, meta-learning approaches like RepMet [7] and Meta RCNN [5] aim to equip models with rapid adaptation

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capabilities, while fine-tuning methods like LSTD [9] and TFA [8] leverage pretrained models for efficient knowledge transfer. In the context of remote sensing images, meta-learning models like FSODM [10] employ episodic training with meta-learning for novel class adaptation, exhibiting a tendency toward overfitting to support sets, potentially limiting generalization. Prototype-CNN [12], utilizing prototypes for efficient class representation, suffers from limitations in capturing intraclass variability and sensitivity to outliers within the support set. SAE-FSDet [46], aiming to mitigate inconsistent labeling assignments, demonstrates a strong reliance on the availability of a sufficient number of novel instances for effective learning, limiting its applicability in data-scarce scenarios. Fine-tuning approaches, such as SIE [47], despite incorporating Gaussian-scale enhancement and multibranch attention, exhibit suboptimal performance due to limited fine-tuning of earlier layers, potentially hindering full adaptation to novel class characteristics. SCoDANet [11], while effective in enhancing feature representation, demonstrates a strong dependence on data complexity and the availability of sufficient novel instances. Insufficient data can impede the learning of robust and discriminative features for novel classes. Finally, UNP [17], despite introducing mechanisms to mitigate negative influences, struggles to effectively isolate confusing negatives, potentially impacting the learning of accurate object representations and generalization. These limitations, such as overfitting, data dependency, and suboptimal fine-tuning, collectively contribute to catastrophic forgetting, where previously learned knowledge of base classes is gradually eroded as the model adapts to novel ones.

Recently, researchers have proposed incremental learning methods to overcome these challenges. Methods like G-FSDet [26] introduce a transfer learning framework for remote sensing that handles similar-looking aerial objects through discriminative loss and prevents forgetting through a dedicated module that separates the learning of old and new knowledge. FSIOD-DFP [27] tackles model forgetting by using frequency-based image prompts to adapt to existing knowledge and separate detection heads for new and previous tasks. However, these approaches, by solely relying on visual features, limit their capacity for creating more powerful representations needed for few-shot learning, resulting in a struggle to maintain stable feature representations when learning from few examples. Recognizing that the problem arises from their limited focus on visual features and the need for representations that are both robust and adaptable, we explore multimodal image–text fusion models, as they offer a more suitable alternative for few-shot adaptation.

B. Multimodal Image–Text Learning

Multimodal image–text learning has explored diverse fusion strategies, broadly categorized into late and multilayer fusion, each exhibiting distinct characteristics. Late fusion processes visual and textual inputs through independent encoders before final combination, employing diverse integration techniques. Within late fusion, contrastive learning methods [48], [49], [50] learn a shared embedding space by contrasting corresponding and noncorresponding image–text pairs, facilitating zero-shot transfer in tasks like image retrieval and cross-modal search. Generative models [51], [52], [53] merge modalities within shared latent spaces for generative tasks such as image captioning and text-to-image synthesis. Fusion-based methods explicitly combine encoded features, employing either direct concatenation [54] or leveraging attention mechanisms [55], [56], [57], [58] within transformer-based architectures for more nuanced fusion. Recently, fusion-based methods [20], [21], [22] have been explored to overcome the problem of data scarcity in FSOD for remote sensing. However, this delayed cross-modal interaction limits its capacity to capture subtle relationships, leading to suboptimal feature alignment, loss of salient cues, and limited hierarchical representation. Multilayer fusion mitigates this by introducing fusion at multiple processing stages, enabling richer cross-modal interaction. For instance, multimodal bottleneck transformer (MBT) [23] used fusion bottlenecks at multiple layers to selectively condense information but faced feature stability challenges. Multimodal end-to-end transformer (METER) [24] aimed to improve information flow to the cross-modal encoder by weighting unimodal representations but degraded performance with pretraining. BridgeTower [25] and ManagerTower [59] employ multiple bridge layers connecting unimodal and cross-modal encoders for more nuanced exchange. Despite the advantage of incorporating textual data with visual modalities, these multimodal approaches still suffer from catastrophic forgetting during few-shot adaptation. During base training, these models learn crucial cross-modal alignments across multiple network layers, mapping both low-level feature patterns (e.g., textures and edges) and high-level semantic concepts between visual and textual modalities. However, when attempting to learn new associations from limited examples, they often disrupt these multilevel feature correspondences, leading to catastrophic forgetting of the base category representations across all feature levels.

C. Catastrophic Forgetting in Multimodal Image–Text Learning

Catastrophic forgetting in image–text models presents a unique challenge, where learned cross-modal relationships deteriorate during adaptation to new tasks [60]. Unlike traditional neural networks, where forgetting primarily affects single-modality features, image–text models face the additional complexity of maintaining aligned visual–textual representations across multiple network layers. Recent works have proposed various approaches to address this challenge, though each faces significant limitations. Early work focused on preserving feature representations through knowledge distillation mechanisms. For instance, Zhang et al. [61] demonstrated that distilling knowledge from multiple teacher models could maintain cross-modal alignments, but this approach requires storing multiple checkpoints and struggles with feature drift. Zhou et al. [62] introduced prototype-guided feature aggregation to stabilize representations, yet their method primarily operates on final-layer features, missing crucial early stage cross-modal relationships that are essential for fine-grained visual understanding. More recent works have explored parameter-efficient adaptation strategies. Methods

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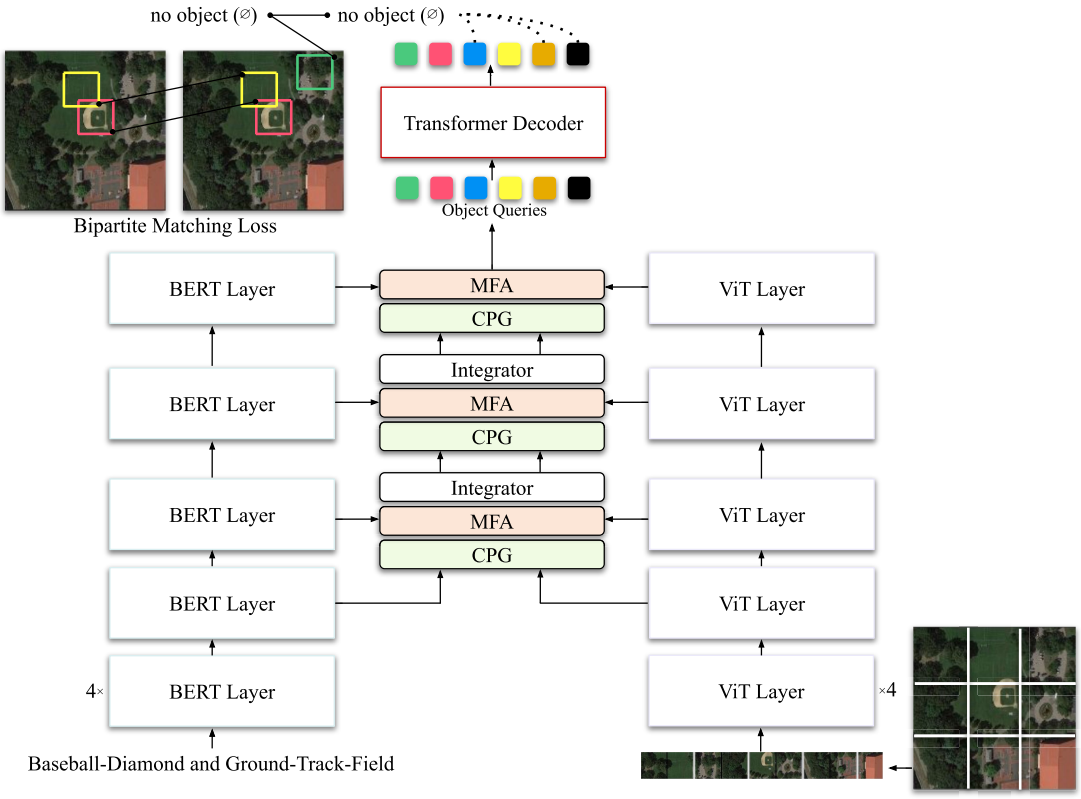


Fig. 3. Protoalign's architecture integrates CPG with hierarchical feature refinement across multiple layers. It uses eight-layer visual and textual encoders based on ViT and BERT, respectively. The last four layers are enhanced with three alignment mechanisms: CPG, which stabilizes features using gated asymmetric attention; MFA, which strengthens semantic relationships through channel-level interactions; and an integrator module, which ensures consistent information flow while preserving modality-specific traits. These refined representations are processed by a DETR decoder for object detection, enabling effective adaptation to novel classes while retaining base knowledge.

like GaB [63] contribute data-free continual learning for visual question answering (VQA) through a balanced, pseudo-rehearsal approach, but its object localization suffers from a lack of explicit spatial training and bias toward textual semantics. Regression-based analytic incremental learning (RAIL) [64] contributes nonforgetting, zero-shot continual learning across domains with a regression adapter, but its object localization is hindered by a textual semantic focus and the adapter's limited spatial processing capabilities. A reliance on implicit pretrained knowledge and a focus on textual semantic rather than spatial aspects during object localization lead to feature space distortion and attention drift.

Protoalign addresses these limitations by incorporating explicit spatial reasoning. Its CPG establishes stable feature anchors that are derived from both visual and textual information, ensuring explicit spatial training. Additionally, Protoalign's multilayer architecture, combined with MFA, directly enhances the network's ability to learn complex spatial relationships, overcoming the limited spatial processing capabilities. By explicitly learning spatial information and integrating it with textual semantics, Protoalign mitigates feature space distortion and attention drift, resulting in a more spatially aware model.

III. Methodology

While existing FSOD methods [20], [21], [22] have explored incorporating text alongside images, they struggle to maintain stable feature representations during few-shot learning, leading to compromised detection performance. This instability stems from the difficulty of preserving learned patterns while incorporating new class knowledge from sparse examples, resulting in catastrophic forgetting. To address these limitations, we propose a prototype-guided multilayer alignment network (Protoalign) as illustrated in Fig. 3 that progressively refines and aligns visual and textual representations across multiple network layers, creating robust features that resist degradation during few-shot adaptation. It processes initial image and text embeddings through a series of prototype-guided alignment layers. Each layer utilizes CPG that establishes stable class-specific feature anchors, MFA module that strengthens semantic relationships through channel-level interactions, and an integrator module that maintains consistent information flow between layers. This hierarchical refinement produces stable cross-modal representations that are then processed by a DETR decoder for object detection. Sections III-A and III-B detail how each component contributes to effective few-shot learning, demonstrating how Protoalign overcomes the limitations of current approaches.

A. Image and Textual Encoding

For visual encoding, we employ Vision Transformer (ViT) [65] to extract hierarchical visual features. Given that an input image I is divided into N_v patches of size $P \times P$, which are linearly projected and combined with position embeddings to

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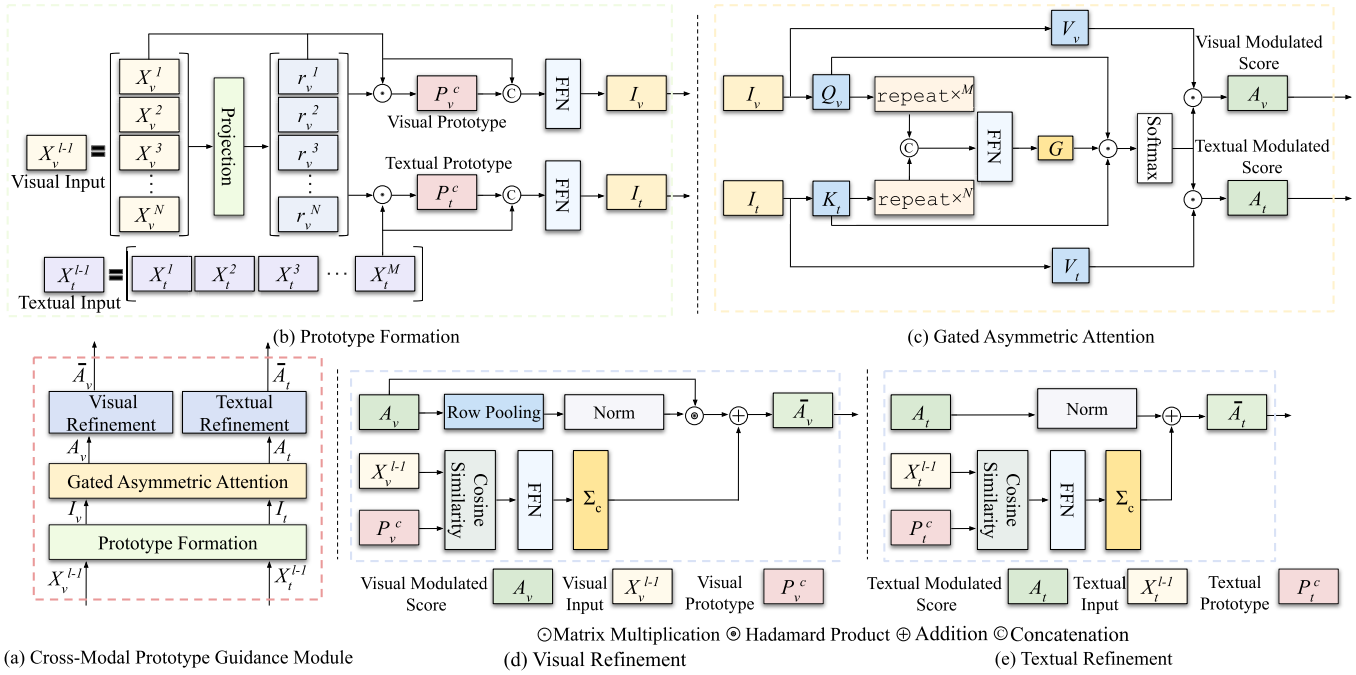


Fig. 4. (a) CPG module stabilizes feature alignments during few-shot adaptation through three stages. (b) Prototype formation, where visual (X_v^{l-1}) and textual (X_t^{l-1}) features generate class-specific prototypes (P_v^c and P_t^c) using reliability scores (r_i) to create refined features (I_v and I_t). (c) Gated asymmetric attention, where visual features query textual ones with gate-modulated attention (G); and feature refinement, where separate (d) visual and (e) textual branches use prototype guidance to produce enhanced class-relevant features ($\bar{A}_v$ and $\bar{A}_t$), ultimately creating representations resistant to feature drift in few-shot scenarios.

create a sequence of visual tokens, these tokens are processed through L_{VIT} transformer layers, producing patch-level representations $X_v = \{v_1, v_2, \ldots, v_N\}$, where v_i represents patch i at layer l. Similarly, for textual encoding, the input text T is first tokenized into a sequence of discrete text tokens. These tokens are embedded into vector representations and combined with position embeddings to encode their sequential order in the text. The embedded tokens are then processed through L_{BERT} transformer layers from bidirectional encoder representations from transformers (BERT) architecture [66], producing contextual word-level features $X_t = t_1, t_2, \ldots, t_N$, where t_i represents the token i's representation that captures both semantic meaning and syntactic relationships within the context of the surrounding text.

B. Prototype-Guided Alignment Layer

1) Cross-Modal Prototype Guidance: CPG, as illustrated in Fig. 4, addresses feature space distortion during few-shot adaptations. CPG is motivated by the need to stabilize feature representations during few-shot learning. It establishes class-specific prototypes that act as anchors in the feature space. These prototypes aggregate discriminative patterns from multiple instances, providing a stable reference point that prevents feature drift during adaptation. This approach ensures that the model maintains decision boundaries for base classes while accommodating novel categories. First, a prototype formation maintains stable feature representations by creating class-specific prototypes that act as anchoring points in the shared embedding space. These prototypes constrain feature drift during adaptation, helping maintain decision boundaries for base classes while accommodating novel categories. Second, a gated asymmetric attention mechanism allows visual features to selectively query relevant textual information, preventing feature drift during adaptation. Finally, a feature refinement stage employs separate visual and textual pathways to enhance learned representations while preserving the distinct characteristics of each modality.

a) Prototype formation: When learning from new data, token-level interactions mechanisms [20], [25] become unstable due to direct feature alignment, leading to degraded detection performance on base classes. To address this, we utilize class-specific prototypes [Fig. 4(b)] that act as reference points that constrain feature drift during adaptation, helping maintain decision boundaries for base classes while accommodating novel categories. These prototypes aggregate discriminative patterns across multiple instances, creating robust class representations that are less sensitive to individual variations or noise.

CPG takes as input visual feature X_v^{l-1} and textual feature X_t^{l-1}. In the first layer, X_v^{l-1} is directly from the ViT, where the image is divided into patches, each processed into a d-dimensional feature vector representing distinct spatial patterns and structural information. Similarly, X_t^{l-1} is directly from BERT, where each token in the text is encoded into a d-dimensional feature vector capturing semantic information. In subsequent layers, these inputs are integrated with visual and textual features from the previous layer's integrator module (Section III-B3). Here, we define each feature vector within X_v^{l-1} or X_t^{l-1} as an instance, representing either a specific visual token or a textual token

$$\begin{aligned} X_v^{l-1} &= \left[X_v^1, \quad X_v^2, \quad \ldots, \quad X_v^N\right], \quad \boldsymbol{X}_v^i \in \mathbb{R}^d \\ X_t^{l-1} &= \left[X_t^1, \quad X_t^2, \quad \ldots, \quad X_t^M\right], \quad \boldsymbol{X}_t^j \in \mathbb{R}^d \end{aligned} \tag{1}$$

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where X_v^i and X_i^j are the visual and textual feature vectors, respectively, and i and j are the visual and textual instances, respectively.

Given the high variability of individual instances within a class in FSOD, stable feature representations are essential. Visual features, which maintain consistent spatial patterns and structural information, provide more reliable cues for prototype formation than text descriptions. During base training, we leverage this insight by computing instance reliability scores from visual features

$$r_i = \sigma\left(W_r X_v^i\right) \tag{2}$$

where σ represents the sigmoid function and W_r denotes learnable parameters. The reliability score r_i reflects the quality of the feature X_v^i.

To determine the class for visual tokens and construct the set S_c, we leverage ground truth annotations through a patch-to-class assignment [67] [see (3)]. Specifically, ViT divides input images into fixed-size nonoverlapping patches, each linearly embedded into a visual token with an exact spatial position determined by grid coordinates. During training, ground truth bounding boxes define regions containing objects of class c, and a visual token is assigned to S_c, if its corresponding patch intersects with any bounding box of class c. During testing, where ground truth annotations are unavailable, the model determines category information through patch-to-cluster assignment [68]. Instead of using spatial overlap with bounding boxes, visual tokens are assigned to classes based on their similarity to the class prototypes learned. Using the class information and reliability scores, we generate class-specific visual and textual prototypes through a weighted average of instance features [see (4)]

$$S_c = \left\{ i \,\middle|\, \text{Patch}\left(X_v^i\right) \cap B_c \neq \emptyset \right\} \tag{3}$$

$$P_v^c = \frac{\sum_{i \in S_c} r_i X_v^i}{\sum_{i \in S_c} r_i}, \quad P_t^c = \frac{\sum_{i \in S_c} r_i X_t^j}{\sum_{i \in S_c} r_i} \tag{4}$$

where P_v^c and P_t^c represent the visual and textual prototypes for class c, respectively; $\text{Patch}(X_v^i)$ represents the spatial region of the ith instance; and B_c is a ground truth bounding box of class c.

Class prototypes distill discriminative patterns from multiple instances, creating class representations that are less sensitive to individual variations. These prototypes, encompassing both visual (P_v^c) and textual (P_t^c) representations, are designed to characterize invariant characteristics of a class by aggregating information across multiple instances. By aggregating features across instances, these prototypes provide stable reference points that constrain feature collapse during adaptation. However, relying solely on class prototypes risks losing instance-specific details, leading to subpar cross-modal alignment. To preserve the fine-grained spatial information and unique characteristics of individual instances while leveraging the generalized knowledge encoded in prototypes, we fuse these class-level prototypes and raw instance features. This fusion is implicitly shaped through gradients flowing from the detector's classification and localization objectives, guiding instance-level feature enhancement as shown in the following equation:

$$\begin{aligned} I_v &= \text{ReLU}\left(W_p^v\left[\text{Concat}\left[X_v^{l-1}, P_v^c\right]\right]\right) \\ I_t &= \text{ReLU}\left(W_p^t\left[\text{Concat}\left[X_t^{l-1}, P_t^c\right]\right]\right) \end{aligned} \tag{5}$$

where projection matrices W_p^v and W_p^t learn to selectively integrate prototype information, strengthening class-relevant patterns while preserving instance-specific details.

During base training, prototypes for the base classes are actively learned and refined, shaped by the incoming data and task objectives through gradient updates based on reliability scores. Once base training is complete, these base class prototypes $\mathcal{P}_{\text{base}}$ become fixed anchors within the prototype space [see (6)]. Their stability is intentionally maintained during the subsequent few-shot adaptation phase; they are not updated with data from novel classes. Instead of modifying existing base prototypes, few-shot adaptation involves computing new prototypes for the novel classes $\mathcal{P}_{\text{novel}}$ using the limited support set data [see (6)]. These newly computed novel prototypes are added to the shared space alongside the fixed base ones [see (7)]. The stability of the base prototypes is further reinforced because they serve as explicit reference points in the updated reliability calculation during adaptation [see (8)], grounding the assessment of new instances and the formation of novel prototypes within the established feature structure

$$\mathcal{P}_{\text{base}} = \left\{ P^c | c \in C_{\text{base}} \right\}$$

$$\mathcal{P}_{\text{novel}} = \left\{ P^k | k \in C_{\text{novel}} \right\} \tag{6}$$

$$\mathcal{P}_{\text{total}} = \mathcal{P}_{\text{base}} \cup \mathcal{P}_{\text{novel}} \tag{7}$$

$$r_i = \sigma\left(W_r\left(X_v^i + \lambda \sum_c \varphi\left(X_v^i, P_v^c\right)\right)\right) \tag{8}$$

where $\varphi = (X \cdot P / \|X\| \cdot \|P\|)$ is the cosine similarity and c and k are the base and novel classes, respectively.

The parameter λ is a learnable parameter that allows the network to automatically balance instance-specific and prototype-based features. Through this implicit learning process, the prototype formation naturally develops statistical efficiency by aggregating instance features, provides regularization through prototype-guided feature enhancement, and maintains stability during adaptation through learned prototype relationships. These properties emerge from the network optimizing its detection performance rather than being explicitly enforced.

b) Gated asymmetric attention: Gated asymmetric attention mechanism [Fig. 4(c)] operates on prototype-integrated visual features ($I_v \in \mathbb{R}^{N \times d_v}$) and textual features ($I_t \in \mathbb{R}^{M \times d_t}$). From these prototype-integrated features, we first project the visual features into visual queries ($Q_v = I_v W_v^Q \in \mathbb{R}^{N \times d_k}$) and visual values ($V_v = I_v W_v^V \in \mathbb{R}^{N \times d_v}$), while the textual features are transformed into textual keys ($K_t = I_t W_t^K \in \mathbb{R}^{M \times d_k}$) and textual values ($V_t = I_t W_t^V \in \mathbb{R}^{M \times d_t}$). The asymmetric design restricts attention flow to a single direction using visual features exclusively as queries (Q_v) to attend to textual features as keys (K_t). This choice is motivated by the understanding that in spatial grounding, visual features should act as the primary reference point, guiding the selection of relevant textual context. Specifically, the visual features after prototype

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integration already contain a rich representation of spatial locations and object boundaries. Therefore, instead of letting the text influence the visual features equally, we want the visual features to guide the attention process by selectively pulling in relevant textual information to augment their existing spatial understanding. This ensures that the visual representation's spatial integrity is maintained, preventing the loss of spatial detail that often occurs with bidirectional attention flows. By having the visual queries drive the attention toward the text keys, we ensure that we gather semantic context like "wing" and "tail" without blurring the aircraft's location as the textual context is only being used to refine the visual understanding and not to reshape it (see the following equation):

$$a = Q_v K_t^{\top}. \tag{9}$$

To establish nonlinear relationships in the dot-product operation, the attention scores (a) are modulated by a learned gate (G). This gate is computed by repeating and concatenating the visual query features (Q_v) (M) times and textual key features (K_t) (N) times, allowing G to capture complex contextual relationships between visual and textual features [see (10)]. Subsequently, G selectively modifies the attention scores (a) based on the combined presence of Q_v and K_t. For instance, in the airport scenario, this enables the model to focus on specific textual features most relevant to the aircraft's appearance and location. By integrating the nonlinearity introduced by G with the spatial preservation of gated asymmetric attention, the model achieves superior alignment and integration of visual and textual information, leading to more accurate identification and localization of objects within the airport scene [see (11)]

$$\begin{aligned} G_q &= \texttt{repeat}(Q_v[:, \texttt{None}, :], M) \\ G_k &= \texttt{repeat}(K_t[\texttt{None}, :, :], N) \\ G &= \sigma\left(W_g\left[\text{Concat}\left[G_q, G_k\right]\right]\right) \qquad (10) \\ \alpha &= \text{softmax}(aG). \qquad (11) \end{aligned}$$

Query tensor (Q_v) and key tensor (K_t) are expanded through the introduction of a unit dimension using the designated `None` notation. This process inserts a new axis with a length of 1 into the tensors. Subsequently, the resulting intermediate tensors undergo replication via a `repeat` function, wherein the tensors are tiled M and N times along the newly inserted unit dimension.

To effectively enhance both visual and textual features, the model employs attention-weighted grounding mechanisms [see (12)]. Specifically, to preserve the spatial structure of visual features (A_v) while enriching them with semantic context, attention-weighted integration is used. Conversely, to enhance semantic representations of textual features (A_t), descriptions are grounded in visual context to capture spatial relationships, like enriching "parallel runways" with observed image details, thus improving localization. This parallel approach ensures that visual features retain their geometric precision and textual features maintain semantic coherence. The modulated attention scores guide separate refinement pathways for each modality

$$\begin{aligned} A_v &= \alpha V_v \\ A_t &= \alpha V_t. \end{aligned} \tag{12}$$

c) Feature refinement: The gated asymmetric attention yields modulated feature associations between visual and textual tokens. While these associations capture instance-specific alignments, class prototypes are introduced as reference points to provide additional structure to the feature space. To effectively combine attention-based associations with prototype guidance, we design separate refinement pathways for visual and textual refinement.

Visual refinement as shown in Fig. 4(d) balances two competing objectives to stabilize spatial feature representations [see (13)]. First, the term row-wise max pooling isolates the strongest correlations between visual regions and textual descriptions. By focusing on the most salient matches, the model prioritizes regions that are discriminative for object detection, such as distinctive edges or textures. However, relying solely on these attention-driven correlations risks overfitting to sparse novel examples, which could distort the feature space of base classes. To counteract this, the prototype similarity term acts as a stabilizer. It measures how closely current visual features align with class-specific prototypes, learned representations of "typical" class patterns from base training. This ensures that even as the model adapts to novel classes, visual features remain tethered to foundational class structures, preventing drastic shifts in decision boundaries. Textual refinement as depicted in Fig. 4(e) anchors semantic representations while preserving contextual richness from visual features. Unlike visual refinement that focuses on spatial locations, textual features retain all attention weights to maintain comprehensive visual associations. This anchoring ensures that when attention captures visual context like "aircraft near runway," the representation stays aligned with consistent class-level semantic patterns [see (14)]

$$\bar{A}_v = \left(\underbrace{\max_j \left(A_{v(i,j)}\right) \odot A_v}_{\text{Attention Highlighting}} + \lambda_v \underbrace{\sum_c w_c \varphi\left(X_v^{l-1}, P_v^c\right)}_{\text{Prototype Anchoring}} \right) \tag{13}$$

$$\bar{A}_t = \lambda_t \underbrace{\sum_c w_c \varphi\left(X_t^{l-1}, P_t^c\right)}_{\text{Prototype Anchoring}} + A_t \tag{14}$$

where w_c is the learnable parameter; $\odot$ is the Hadamard product; λ_v, λ_t, and φ calculate the feature-prototype cosine similarity; and λ_t and λ_v learnable parameter controls the tradeoff between adaptability and stability.

2) Multimodal Feature Aggregation: While CPG focuses on instance-level feature alignment, these instance-level alignments can be fragile in few-shot scenarios due to individual variations. MFA, as illustrated in Fig. 5(a), seeks to capture more stable cross-modal patterns by focusing on channel-level interactions. It identifies systematic coactivation patterns between visual and textual channels, which remain consistent across different examples. This approach provides a broader structural understanding that complements instance-level alignments. First, MFA captures systematic coactivation patterns between modalities, for example, visual channels encoding specific edge patterns consistently activate alongside

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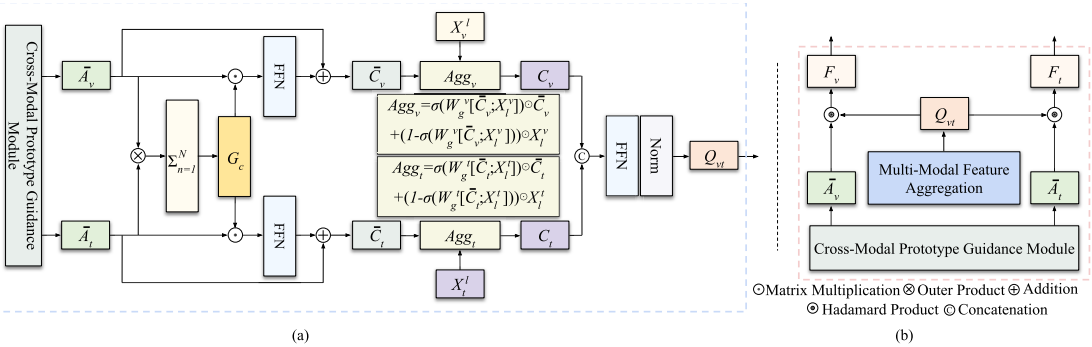


Fig. 5. (a) MFA enhances cross-modal relationships by generating a channel correlation matrix G_c from $\bar{A}_v$ and $\bar{A}_t$, capturing coactivation patterns between visual and textual features from CPG. Refined features $\bar{C}_v$ and $\bar{C}_t$ integrate with next-layer features X_l^v and X_l^t for adaptive enhancement across network depths. (b) Integrator combines a unified representation Q_{vt} from MFA with refined CPG features through element-wise multiplication, producing F_v and F_t, effectively balancing attention with structural correspondences while preserving modality-specific characteristics.

textual channels representing boundary-related concepts ("contour," "outline," and "border"). This channel-level consistency helps prevent attention drift by maintaining stable cross-modal activation patterns even as the model adapts to new classes. These channel-level relationships are then enhanced using features from the next layer (X_l^v and X_l^t), allowing the network to adapt and refine these correspondences hierarchically. This approach differs from instance-level matching, as it identifies robust cross-modal patterns, and structural correspondences, that generalize across examples and remain consistent across network depths, making it particularly valuable in few-shot learning scenarios where robust feature mappings are needed to avoid overfitting. Such structural correspondences, operating at the channel level instead of the instance level, provide a more stable learning mechanism that relies on consistent relationships within and across modalities.

Given the enhanced features $\bar{A}_v$ and $\bar{A}_t$ from CPG, we model channel interactions through token-wise outer products [see (15)]. The token-wise outer product effectively captures comprehensive pairwise relationships between modalities because it ensures that every element from one modality interacts with every element from the other modality, creating a complete correlation matrix that preserves both intramodal and intermodal dependencies. Unlike inner product operations that require index matching and selective multiplication, the outer product eliminates the need for index matching as each pair of nonzero elements from the visual and textual features produces meaningful outputs [69]. This interaction pattern is essential for overcoming the heterogeneous modality gap, as it allows the network to identify how each visual feature responds in relation to each text feature, yielding a channel correlation matrix G_c

$$G_c = \frac{1}{N}\sum_{n=1}^{N}\left(\bar{A}_v \otimes \bar{A}_t\right) \tag{15}$$

where $\otimes$ indicates the outer product.

The transformation of features from one modality, based on the channel-level correlation matrix derived from the other, injects cross-modal context and refines features based on their intermodality alignment at the channel level. Channel-level operations consider the relationships and correlations between channels while preserving the spatial structure within each channel. This approach allows the network to model global interactions between visual and textual modalities by weighting or combining channels differently, without disrupting the spatial arrangement within individual feature maps preserving the spatial details for accurate object localization. This selective adjustment amplifies strongly correlated channel-level features while de-emphasizing weakly correlated ones, leading to a shared representational space where modalities are aligned [see (16) and (17)]. This channel-aware mechanism helps bridge the semantic gap between the modalities, ultimately facilitating the learning of joint representations that enable the model to reason across modalities effectively

$$\bar{C}_v = \mathrm{LN}\left(W_v^1 \mathrm{ReLU}\left(W_v^2\left(G_c \bar{A}_t\right)\right) + \bar{A}_v\right) \tag{16}$$

$$\bar{C}_t = \mathrm{LN}\left(W_t^1 \mathrm{ReLU}\left(W_t^2\left(G_c^T \bar{A}_v\right)\right) + \bar{A}_t\right) \tag{17}$$

where LN represents the layernorm and W_v^2, W_v^1, W_t^1, and W_t^2 are projection matrices.

MFA creates hierarchical feature connections by fusing correlation-enhanced features from the current layer ($\bar{C}_v$ and $\bar{C}_t$) with encoder features from the next layer (X_l^v and X_l^t). This hierarchical integration strategy enables progressive refinement of cross-modal representations by leveraging both fine-grained correlations from current layers and high-level semantic features from deeper layers. Through a learnable gating mechanism [see (18)], this integration dynamically balances the cross-modal patterns captured by channel correlations with modality-specific features learned by the encoders. When the encoded features contain strong discriminative patterns, the gates allow more information from X_l^v and X_l^t. Conversely, when cross-modal relationships are crucial, the gates emphasize the correlation-enhanced features $\bar{C}_v$ and $\bar{C}_t$. This adaptive balance is computed as

$$G_v = \sigma\left(W_g\left[\mathrm{Concat}\left[\bar{C}_v, X_l^v\right]\right]\right)$$

$$G_t = \sigma\left(W_g^t\left[\mathrm{Concat}\left[\bar{C}_t, X_l^t\right]\right]\right) \tag{18}$$

$$C_v = G_v \odot \bar{C}_v + (1 - G_v) \odot X_l^v \tag{19}$$

$$C_t = G_t \odot \bar{C}_t + (1 - G_t) \odot X_l^t. \tag{20}$$

The integration mechanism adaptively weights the contribution of enhanced and layer-specific features, optimizing the

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TABLE I
BASE-NOVEL SPLIT FOR REMOTE SENSING DATASETS IN FSOD

Dataset	Split	Novel					Base
iSAID[30]	1	helicopter	ship	plane	large vehicle		rest
	2	baseball diamond	soccer ball field	roundabout			rest
	3	ground track field basketball court	helicopter	baseball diamond	roundabout	soccer ball field	rest
DIOR[31]	1	baseball field	basketball court	bridge	chimney	ship	rest
	2	airplane	airport	expressway toll station	harbor	ground track field	rest
	3	dam	golf course	storage tank	tennis court	vehicle	rest
	4	express service area	overpass	stadium	train station	windmill	rest
FAIR1M-Airplane [32]	1	a3350	boeing737	boeing747			rest
NWPU VHR-10[3]	1	airplane	baseball diamond	tennis court			rest
	2	basketball	ground track field	vehicle			rest

balance between cross-modal alignment and modality-specific characteristics. Finally, the enhanced representations combine through modality-specific weights that adapt to their relative discriminative power as shown in the following equation:

$$Q_{vt} = \mathrm{FFN}([\mathrm{Concat}[C_v, C_t]]). \tag{21}$$

3) Integrator: The integrator as depicted in Fig. 5(b) combines the unified representation Q_{vt} from MFA with the refined attention features $\bar{A}_v$ and $\bar{A}t$ from CPG, facilitating effective information propagation into subsequent layers. This integration is designed to combine both instance-level feature correlations Q_{vt} and channel-level correlations $\bar{A}_v$ and $\bar{A}_t$ through element-wise multiplication. This approach serves as an effective cross-modal fusion strategy because it enables selective feature modulation where one modality can act as attention weights to emphasize or suppress features from another modality [70]. This is calculated as follows:

$$F_v = Q_{vt} \odot \bar{A}_v \tag{22}$$

$$F_t = Q_{vt} \odot \bar{A}_t \tag{23}$$

where F_t and F_v are the integrated textual and visual representations, respectively.

These integrated representations, F_t and F_v, are then passed as the inputs X_t^{l-1} and X_v^{l-1} to the CPG module in the subsequent layer of the network, enabling an iterative fusion. However, in the final layer, the integrator is removed, and the unified representation Q_{vt} from MFA is directly fed into the decoder for object detection, as shown in Fig. 3.

C. Decoder

Following the single-stage detection approach of DETR [38] and unified multimodal fusion transformer (UMFT) [20], we formulate object detection as a direct set prediction problem. We use a global set-based loss function combined with bipartite matching and employ a transformer encoder–decoder architecture to generate unique predictions. The ground truth objects are represented by y, while the predicted set of objects is denoted as $\hat{y} = \{\hat{y}_i\}_{i=1}^{N}$. Our objective is to minimize the set-based loss by finding a permutation $\sigma \in \mathcal{L}_N$ of N elements that results in the lowest cost (see the following equation):

$$\hat{\sigma} = \arg\min_{\sigma \in \Phi^N} \sum_{i=1}^{N} L_{\mathrm{match}}\big(y_i, \hat{y}_\sigma(i)\big) \tag{24}$$

where $L_{\mathrm{match}}(y_i, \hat{y}_\sigma(i))$ represents the pairwise matching cost between the ground truth y_i and the corresponding prediction $\hat{y}_\sigma(i)$.

In the decoder stage, positional embeddings are incorporated into multiple transformer decoder layers, allowing the model to simultaneously identify N objects at each layer. The model is trained using a combination of negative log-likelihood loss for class and bounding box regression loss

$$L_v\big(y, \hat{y}\big) = \sum_{i=1}^{N} \big[-\log \hat{p}_{\hat{\sigma}(i)}(c_i)\big] + L_{\mathrm{box}}\big(b_i, \hat{b}_{\hat{\sigma}}(i)\big) \tag{25}$$

where $\hat{p}_{\hat{\sigma}(c_i)}$ represents the classes and $L_{\mathrm{box}}(b_i, \hat{b}_{\hat{\sigma}}(i))$ denotes the normalized loss for bounding box regression.

IV. EXPERIMENTATION

A. Dataset

iSAID dataset is a large-scale aerial imagery collection featuring both instance-level mask and fine-grained bounding box annotations, comprising 2806 high-resolution images with 655 451 object instances across 15 classes. The classes include various infrastructure, transportation, and urban elements (ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, bridge, large vehicle, small vehicle, helicopter, swimming pool, roundabout, soccer ball field, plane, and harbor). The imagery is sourced from diverse locations worldwide with varying resolutions. Following the splits defined in Table I, we evaluate FSOD with $k = 10$, 50, 100 shots.

DIOR dataset contains 23 463 Google Earth images with 192 472 object instances spanning 20 classes, encompassing various transportation, infrastructure, and sports facilities (airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, harbor, golf course, ground track field, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill). The imagery has spatial resolutions between 0.5 and 30 m. Following the splits defined in Table I, we evaluate FSOD with $k = 3$, 5, 10, 20 shots.

FAIR1M dataset is a comprehensive remote sensing image dataset containing over 15 000 images and 1.02 million object instances, with image resolutions of 0.3–0.8 m and sizes ranging from 1000 to 10 000 pixels. In our experiments, we focus on FAIR1M-Airplane, a subset containing 11 aircraft

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subcategories, including various Boeing and Airbus models, COMAC aircraft, and others. Following prior work [27], we designate three classes as novel and eight as base classes as shown in Table I, converting rotated annotations to horizontal ones for FSOD evaluation with $k = 2, 3, 5, 10$ shots.

NWPU VHR-10 dataset consists of 800 high-resolution images (715 RGB from Google Earth and 85 color infrared from Vaihingen), containing 3775 object instances across 10 classes in 650 images, plus 150 negative sample images. The classes include various infrastructure and transportation objects (airplane, baseball diamond, basketball court, bridge, harbor, ground track field, ship, storage tank, tennis court, and vehicle). Following the splits shown in Table I, we evaluate FSOD performance with $k = 3, 5, 10, 20$ shots.

B. Implementation Details and Metrics

Protoalign consists of two primary encoding components: an eight-layer BERT-base textual encoder ($L_{\text{BERT}} = 8$) for processing textual descriptions and an eight-layer CLIP-ViT-B/16 vision encoder ($L_{\text{VIT}} = 8$) for extracting visual features. The last four layers of both encoders ($L_{\text{Protoalign}} = 4$) are connected to our proposed multilevel alignment architecture. Specifically, at each of these layers, the CPG module creates class-specific prototypes with a feature dimension of 512, while the MFA module processes channel-level interactions using a hidden dimension of 256. The integrator combines these enhanced representations to enable the iterative information flow. The enhanced representations are then processed by a DETR decoder with six decoder layers and eight attention heads for object detection. During base training, we use the adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1 \times 10^{-8}$, starting with a learning rate of 1×10^{-4} that decays by a factor of 0.1 at 30 000, 40 000, and 45 000 iterations with weight decay being set to 1×10^{-4}. For few-shot fine-tuning, the initial learning rate is increased to 5×10^{-4} with a cosine warmup during the first 1000 iterations, followed by step decays at 1000, 3000, and 4000 iterations. The weight decay is reduced to 1×10^{-5} during fine-tuning. All input images are resized to 512×512 pixels and normalized using ImageNet statistics (mean = [0.485, 0.456, 0.406] and std = [0.229, 0.224, 0.225]). During training, we apply standard data augmentation techniques, including random horizontal flipping (probability = 0.5), random resizing (scale range [0.8, 1.2]), and random cropping. For the text encoder, we use a maximum sequence length of 77 tokens, with special tokens (CLS) and (SEP) added for BERT encoding. For optimization stability, we employ gradient clipping with a maximum norm of 0.1 and layer-wise learning rate decay where the learning rate for the encoder layers decreases by a factor of 0.9 for each layer from top to bottom. The prototype formation in CPG uses momentum update with $\alpha = 0.999$ for stable prototype learning. Performance evaluation uses mean average precision (mAP) with IoU thresholds of 0.5.

C. Ablation Studies

To validate our design choices and understand the contribution of each architectural component, we conduct comprehensive ablation studies focusing on seven critical aspects. First, we investigate how progressive feature integration strategies affect cross-modal learning (Section IV-C1), as this is crucial for understanding early versus late fusion impacts. We then analyze how individual component (Fig. 6) training examples through shot-wise performance scaling (Section IV-C2). Our novel gated asymmetric attention mechanism is evaluated against established approaches [25], [71], [72] (Section IV-C3). We further assess robustness to textual variations, including paraphrasing and noise (Section IV-C4). We investigate feature hierarchy through systematic replacement experiments (Section IV-C5). Moreover, we analyze feature selection's impact on efficiency–performance tradeoffs (Section IV-C6) and impact of prototype on feature retention (Section IV-C7).

All experiments for first six ablations use the challenging split 3 of iSAID as shown in Table I with standard few-shot settings (10, 50, and 100 shots), evaluating both performance (mAP on base/novel/all classes) and efficiency [parameters/floating point operations (FLOPs)]. Additionally, we conduct a separate analysis on split 1 to study the impact of prototypes on feature retention (Section IV-C7). Split 1 was chosen because it exhibits a greater likelihood of catastrophic forgetting, caused by feature similarity between base and novel classes. This characteristic makes it advantageous for both visualizing and analyzing feature retention within a few-shot learning context.

1) Impact of Progressive Cross-Modal Feature Integration: The effectiveness of image–text models critically depends on the network layer at which cross-modal feature integration occurs. Early fusion can capture fine-grained correlations but risks mixing immature features, while late fusion, while operating on more developed semantic representations, may miss crucial low-level relationships. To investigate this tradeoff, we analyze fusion strategies ranging from early (eight-layer) to late-stage (three-layer) integration [Table II and Fig. 7(a)]. A baseline late fusion (three-layer) shows limited cross-modal interaction ($0.62 \rightarrow 0.66 \rightarrow 0.67$ AP base and $0.12 \rightarrow 0.28 \rightarrow 0.31$ AP novel), and very early fusion (eight-layer) shows better base performance (0.76 AP and 0.36 AP novel at 100 shots), a 13.4% relative improvement over the baseline. This improvement implies early integration aids in building transferable features but at a substantial computational cost (322 M parameters and 472 GFLOPs). Intermediate fusion (four-layer) emerges as a strong tradeoff, achieving good performance ($0.70 \rightarrow 0.70 \rightarrow 0.72$ AP base and $0.16 \rightarrow 0.33 \rightarrow 0.35$ AP novel) with reasonable computational requirements (158 M parameters and 278 GFLOPs), suggesting that allowing features to mature before integration is an effective approach.

2) Impact of Architectural Components on Shot-Wise Performance: The ablation study shown in Table III and Fig. 7(b) evaluates the architectural components' contribution to mitigating catastrophic forgetting during few-shot adaptation. Baseline performance shows limited shot-wise scalability, cross-attention [71] has marginal improvements in novel class detection ($0.05 \rightarrow 0.08 \rightarrow 0.10$ AP), and base class performance saturates ($0.42 \rightarrow 0.44 \rightarrow 0.44$ AP), while bridge layer [25] shows similar constraints ($0.05 \rightarrow 0.10$

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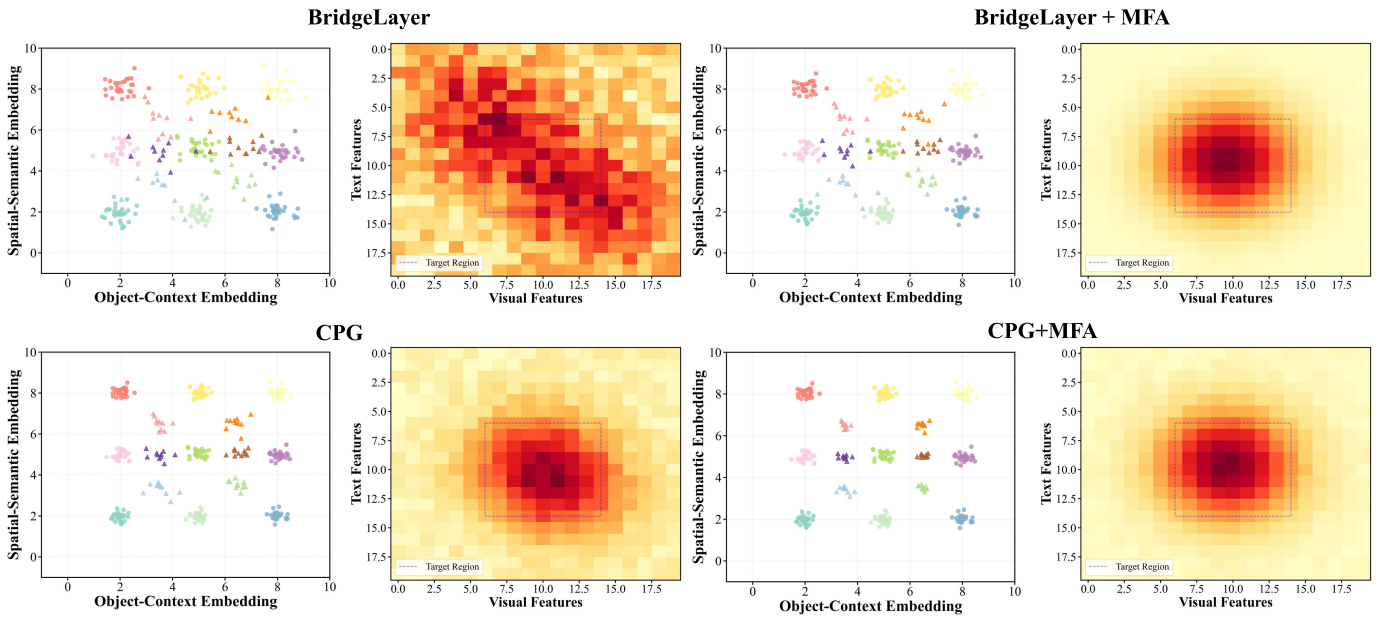


Fig. 6. Cross-modal feature alignment visualization. (Left) t-SNE scatter plots show instance distributions from layer 8. (Right) Density heatmaps display normalized pairwise similarities between aligned features, revealing modality alignment strength. Results from iSAID split-3 at a 100-shot setting.

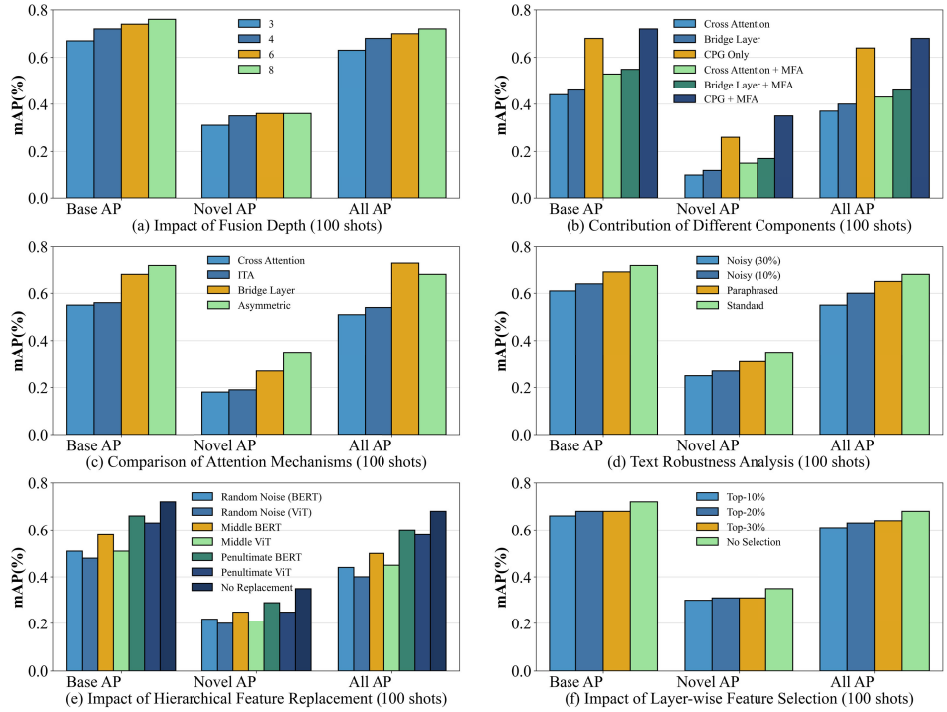


Fig. 7. Ablation studies of protoalign on iSAID split 3 under 100-shot setting. (a) Impact of fusion depth showing optimal performance at four layers with diminishing returns beyond. (b) Analysis of architectural components. (c) Comparison of attention mechanisms. (d) Text robustness analysis. (e) Hierarchical feature replacement experiments. (f) Layer-wise feature selection. All experiments measure base AP (base classes), novel AP (novel classes), and all AP (overall performance).

$\rightarrow$ 0.12 AP novel and 0.44 $\rightarrow$ 0.47 $\rightarrow$ 0.46 AP base). CPG alone significantly improves shot-wise adaptability, with substantial gains in both novel (0.11 $\rightarrow$ 0.20 $\rightarrow$ 0.26 AP) and base class detection (0.61 $\rightarrow$ 0.64 $\rightarrow$ 0.68 AP). MFA with baseline architectures shows that complementary mechanisms are important. Specifically, MFA enhances the performance compared to cross attention (0.51 $\rightarrow$ 0.52 $\rightarrow$ 0.53 AP base) and consistently maintains the performance compared with bridge layer (0.55 $\rightarrow$ 0.55 $\rightarrow$ 0.55 AP). Combining CPG and MFA achieves optimal performance scaling with robust improvement in novel class detection (0.16 $\rightarrow$ 0.33 $\rightarrow$ 0.35 AP) while maintaining base class performance (0.70 $\rightarrow$ 0.70 $\rightarrow$ 0.72 AP), indicating effective mitigation of catastrophic forgetting through complementary mechanisms. These empirical results validate the architecture's effectiveness in handling the challenges of FSOD.

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TABLE II

IMPACT OF LAYER DEPTH ON MODEL PERFORMANCE WITH MAP@0.5/MAP@0.5:0.95

Layers	10 Shots			50 Shots			100 Shots			Parameters	Flops	Training	Inference
	Base	Novel	All	Base	Novel	All	Base	Novel	All	(M)	(G)	(Hours)	(ms)
3	0.62/0.45	0.12/0.09	0.60/0.46	0.66/0.49	0.28/0.20	0.64/0.49	0.67/0.53	0.31/0.22	0.63/0.49	122	210	10.6	45.2
4	0.70/0.51	0.16/0.11	0.63/0.48	0.70/0.52	0.33/0.24	0.66/0.51	0.72/0.56	0.35/0.26	0.68/0.54	158	278	12.6	69.8
6	0.70/0.50	0.16/0.10	0.64/0.47	0.70/0.50	0.34/0.24	0.67/0.51	0.74/0.57	0.36/0.27	0.70/0.55	261	322	19.2	104.8
8	0.72/0.52	0.17/0.12	0.65/0.48	0.72/0.54	0.34/0.24	0.68/0.52	0.76/0.59	0.36/0.27	0.72/0.58	322	472	22.3	146.3

TABLE III

IMPACT OF MODULES ON PERFORMANCE WITH MAP@0.5/MAP@0.5:0.95

Baseline		Protoalign Layer		10 Shots			50 Shots			100 Shots			Parameters	Flops	Training	Inference
Cross Attention	Bridge Layer	CPG	MFA	Base	Novel	All	Base	Novel	All	Base	Novel	All	(M)	(G)	(Hours)	(ms)
✓	×	×	×	0.42/0.32	0.05/0.04	0.35/0.27	0.44/0.33	0.08/0.06	0.36/0.28	0.44/0.33	0.10/0.07	0.37/0.28	112	188	8.5	40.4
×	✓	×	×	0.44/0.34	0.05/0.04	0.37/0.28	0.47/0.35	0.10/0.07	0.39/0.30	0.46/0.36	0.12/0.09	0.40/0.31	240	290	14.8	90.7
×	×	✓	×	0.61/0.45	0.11/0.08	0.55/0.40	0.64/0.49	0.20/0.15	0.58/0.44	0.68/0.51	0.26/0.19	0.64/0.48	118	204	10.2	53.8
✓	×	×	✓	0.51/0.39	0.07/0.05	0.41/0.31	0.52/0.38	0.16/0.12	0.42/0.32	0.53/0.41	0.15/0.11	0.43/0.33	143	251	10.7	64.3
×	✓	×	✓	0.55/0.42	0.09/0.07	0.44/0.33	0.55/0.41	0.17/0.13	0.44/0.34	0.55/0.41	0.17/0.13	0.46/0.36	297	348	20.2	110.5
×	×	✓	✓	0.70/0.51	0.16/0.11	0.63/0.48	0.70/0.52	0.33/0.24	0.66/0.51	0.72/0.56	0.35/0.26	0.68/0.54	158	278	12.6	69.8

TABLE IV

IMPACT OF ATTENTION MECHANISM IN CPG. CA: CROSS ATTENTION, BL: BRIDGE LAYER, AND GAA: GATED ASYMMETRIC ATTENTION

Alignment	10 Shots			50 Shots			100 Shots			Parameters	Flops
	Base	Novel	All	Base	Novel	All	Base	Novel	All	(M)	(G)
CA [74]	0.50	0.08	0.42	0.57	0.16	0.47	0.55	0.18	0.51	166	300
ITA [73]	0.48	0.09	0.42	0.59	0.18	0.48	0.56	0.19	0.54	141	249
BL [25]	0.52	0.11	0.48	0.62	0.25	0.54	0.68	0.27	0.73	194	312
GAA	0.70	0.16	0.63	0.70	0.33	0.66	0.72	0.35	0.68	158	278

3) Analysis of Attention Mechanisms in Cross-Modal Learning: Effective FSOD requires cross-modal alignment that preserves modality-specific characteristics. Bidirectional cross-modal fusion can disrupt the spatial relationships in visual features, while textual features require selective integration for semantic coherence. To address this, we propose a gated asymmetric attention mechanism, which we evaluate against established approaches [Table IV and Fig. 7(c)]. Our analysis reveals limitations in symmetric approaches. Cross-attention's bidirectional feature mixing disrupts spatial coherence, leading to inconsistent base class performance (0.50 → 0.57 → 0.55 AP) and limited novel class improvement (0.08 → 0.16 → 0.18 AP). ITA shows similar limitations (0.09 → 0.18 → 0.19 AP novel). Bridge layer has better novel class performance (0.11 → 0.25 → 0.27 AP) but incurs a high computational cost (194 M parameters and 312 GFLOPs). Our gated asymmetric attention mechanism addresses these issues by restricting information flow to visual features querying textual information, which preserves spatial structure while enriching visual representations with semantic context. This enables superior performance scaling (0.70 → 0.70 → 0.72 AP base and 0.16 → 0.33 → 0.35 AP novel) with lower computational requirements (158 M parameters and 278 GFLOPs), proving effective for novel class generalization through maintaining visual structure and incorporating semantic guidance.

4) Analysis of Cross-Modal Robustness Against Text Variations: Real-world text descriptions often have variations, from simple paraphrases (e.g., "airplane on runway" → "aircraft during takeoff") to noisy inputs (e.g., "airplne on runwy"). The effectiveness of cross-modal learning depends on maintaining semantic alignment despite these perturbations. We evaluate

TABLE V

TEXT ROBUSTNESS AND GENERALIZATION. NC10: NOISY CAPTION 10%, NC30: NOISY CAPTION 30%, PC: PARAPHRASED CAPTION, AND SC: STANDARD CAPTIONS

Captions	10 Shots			50 Shots			100 Shots		
	Base	Novel	All	Base	Novel	All	Base	Novel	All
NC30%	0.60	0.10	0.54	0.57	0.19	0.44	0.61	0.25	0.55
NC10%	0.62	0.12	0.55	0.59	0.23	0.49	0.64	0.27	0.60
PC	0.64	0.13	0.60	0.66	0.28	0.62	0.69	0.31	0.65
SC	0.70	0.16	0.63	0.70	0.33	0.66	0.72	0.35	0.68

TABLE VI

IMPACT OF HIERARCHICAL FEATURE REPLACEMENT IN MFA. FB: FINAL BERT, MB: MEDIUM BERT, FV: FINAL VIT, MV: MEDIUM VIT, PB: PENULTIMATE BERT, PV: PENULTIMATE VIT, AND RN: RANDOM NOISE

Model	10 Shot			50 Shots			100 Shots		
	Base	Novel	All	Base	Novel	All	Base	Novel	All
MFA (FB → RN)	0.48	0.07	0.49	0.50	0.19	0.48	0.51	0.22	0.44
MFA (FV → RN)	0.40	0.04	0.44	0.46	0.22	0.44	0.48	0.20	0.40
MFA (FB → MB)	0.57	0.10	0.55	0.57	0.24	0.54	0.58	0.25	0.50
MFA (FV → MV)	0.52	0.08	0.49	0.52	0.25	0.50	0.51	0.21	0.45
MFA (FB → PB)	0.62	0.12	0.59	0.66	0.30	0.61	0.66	0.29	0.60
MFA (FV→ PV)	0.58	0.09	0.53	0.61	0.27	0.53	0.63	0.25	0.58
MFA (N/A)	0.70	0.16	0.63	0.70	0.33	0.66	0.72	0.35	0.68

TABLE VII

IMPACT OF LAYER-WISE FEATURE SELECTION IN MFA

Model	10 Shot			50 Shots			100 Shots		
	Base	Novel	All	Base	Novel	All	Base	Novel	All
MFA (Top-10%)	0.62	0.11	0.59	0.64	0.27	0.60	0.66	0.30	0.61
MFA (Top-20%)	0.62	0.12	0.60	0.66	0.29	0.61	0.68	0.31	0.63
MFA (Top-30%)	0.64	0.12	0.60	0.67	0.30	0.62	0.68	0.31	0.64
MFA (No)	0.67	0.16	0.63	0.70	0.33	0.66	0.72	0.35	0.68

Protoalign's robustness using character-level noise injection and paraphrases [Table V and Fig. 7(d)]. Our analysis reveals three key insights. First, Protoalign exhibits remarkable noise resilience, maintaining performance even with 30% character corruption (0.60 → 0.57 → 0.61 AP base and 0.10 → 0.19 → 0.25 AP novel) due to prototype-guided learning anchoring features. Second, Protoalign effectively handles semantic variations, as demonstrated by its paraphrase performance (0.64 → 0.66 → 0.69 AP base and 0.13 → 0.28 → 0.31 AP novel), indicating semantic understanding rather than pattern matching.

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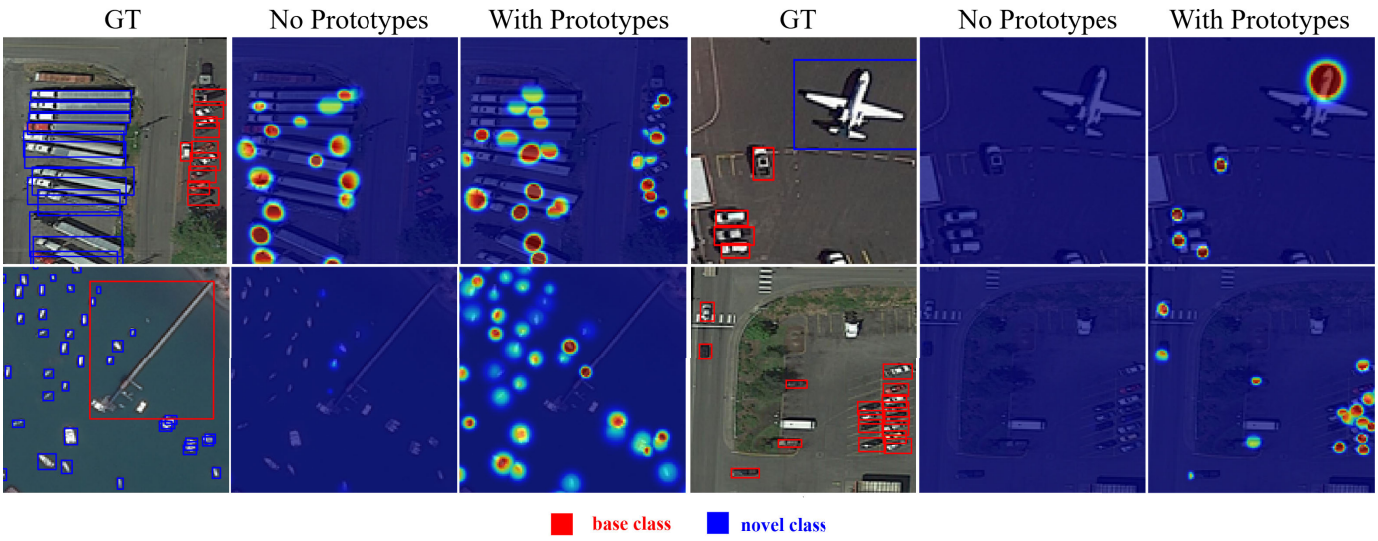


Fig. 8. Feature map visualizations demonstrating the impact of prototype guidance on FSOD.

TABLE VIII
IMPACT OF PROTOTYPES ON RETENTION ON iSAID SPLIT 1. NP: NO PROTOTYPES AND WP: WITH PROTOTYPE

Setting	10 Shot			50 Shots			100 Shots		
	Base	Novel	All	Base	Novel	All	Base	Novel	All
NP	0.62	0.12	0.53	0.59	0.22	0.53	0.60	0.32	0.52
WP	0.70	0.16	0.63	0.70	0.33	0.66	0.72	0.35	0.68

5) Analysis of Hierarchical Feature Contributions in MFA: Image–text models demonstrate hierarchical feature learning, with early layers encoding basic patterns (e.g., edges and textures) and deeper layers representing high-level concepts. To quantify this impact on few-shot detection, we conduct feature replacement experiments [Table VI and Fig. 7(e)]. Our analysis reveals three key insights about MFA. First, random noise replacement shows that visual features are more critical for cross-modal alignment, with ViT noise causing steeper degradation (0.40 → 0.46 → 0.48 AP base) than BERT noise (0.48 → 0.50 → 0.51 AP base) and a more significant impact on novel class detection (0.04 AP versus 0.07 AP at ten shots), suggesting that visual features provide stronger anchors. Second, layer replacement shows progressive semantic abstraction, with penultimate layer features showing substantial base class improvement (0.62 → 0.66 → 0.66 AP). Each layer contributes approximately 0.05 AP, suggesting structured information accumulation. The smaller gap between the penultimate and final layers (0.72 versus 0.66 AP base at 100 shots) suggests nearing semantic saturation in deeper layers. Third, MFA (with no feature replacement) shows superior performance (0.70 → 0.70 → 0.72 AP base and 0.16 → 0.33 → 0.35 AP novel), validating the importance of preserving feature hierarchies. Effective few-shot learning requires both fine-grained early patterns and abstract deeper features, as low-level visual details distinguish subtle object differences, while high-level semantic features enable generalization.

6) Analysis of Feature Selection Impact on MFA: The relationship between feature sparsity and model generalization raises the question of whether selective feature utilization can maintain cross-modal understanding while improving efficiency, which is especially relevant in few-shot scenarios. We investigate this by conducting feature selection experiments using importance-based pruning [Table VII and Fig. 7(f)]. Our analysis reveals that MFA is resilient to feature pruning, and even with only the top 10% of features (based on attention weights), the model maintains robust performance (0.62 → 0.64 → 0.66 AP base and 0.11 → 0.27 → 0.30 AP novel), suggesting that core semantic alignments are captured by few features, which likely correspond to discriminative patterns and descriptions. Increasing feature retention provides diminishing returns (+0.02 AP novel at 100 shots from 10% to 20% retention and negligible gain from 20% to 30%), indicating a feature importance hierarchy. However, MFA (no feature selection) shows significantly better performance (0.67 → 0.70 → 0.72 AP base and 0.16 → 0.33 → 0.35 AP novel), particularly for novel classes (0.35 versus 0.31 AP at 100 shots), suggesting that the full feature spectrum provides fine-grained distinctions important for generalization, with seemingly redundant features encoding subtle variations crucial for adapting to unseen categories.

7) Impact of Prototypes on Catastrophic Forgetting: Table VIII and Fig. 8 demonstrate that prototype guidance is crucial for maintaining feature stability in few-shot adaptation scenarios. Feature map visualizations demonstrate that prototypes facilitate superior feature organization and class separation. The vehicle scene showed more defined decision boundaries with prototypes than without, while ships exhibited more localized activations along the harbor when prototypes were included. When prototypes were removed, classification performance dropped significantly by −43% and −20% in the base and novel classes, respectively, confirming the effect of prototype guidance in mitigating catastrophic forgetting. Collectively, these findings confirm that prototype-guided learning improves feature coherence, promotes greater knowledge retention, and enhances novel class adaptation.

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TABLE IX

IMPACT OF FUSION MECHANISM IN INTEGRATOR ON iSAID SPLIT 1. EF: ELEMENT WISE FUSION, AW: ATTENTION-BASED WEIGHTING, AND CFA: CROSS-MODAL FUSION ADAPTER

Method	10 Shot			50 Shots			Parameters	FLOPS	Convergence
	Base	Novel	All	Base	Novel	All	(M)	(G)	Epochs
EF	0.70	0.16	0.63	0.70	0.33	0.66	0	2.4	26
AW	0.70	0.17	0.63	0.71	0.33	0.67	4.4	9.8	56
CFA	0.71	0.18	0.64	0.72	0.33	0.68	6.9	15.58	70

TABLE X

FSOD PERFORMANCE ON iSAID UNDER (10-, 50-, AND 100-SHOT SETTINGS) ON THREE DIFFERENT BASE/NOVEL SPLITS. THE BOLD DENOTES THE BEST RESULTS, AND UNDERLINE DENOTES THE SECOND BEST

Split	Methods	10-shot			50-shot			100-shot		
		Base	Novel	All	Base	Novel	All	Base	Novel	All
1	RAIL [65]	0.36	0.06	0.30	0.37	0.09	0.32	0.37	0.11	0.34
	TFA [8]	0.59	0.03	0.48	0.48	0.05	0.50	0.61	0.05	0.50
	iMTFA [75]	0.57	0.04	0.49	0.58	0.10	0.50	0.59	0.12	0.52
	METER [24]	0.60	0.06	0.52	0.58	0.13	0.50	0.59	0.15	0.52
	BridgeTower [25]	0.60	0.07	0.52	0.61	0.15	0.53	0.58	0.15	0.50
	DH-FSDet [76]	0.65	0.05	0.52	0.65	0.13	0.53	0.65	0.17	0.54
	UMFT [20]	0.63	0.08	0.53	0.62	0.20	0.54	0.62	0.32	0.55
	G-FSDet [26]	0.64	0.09	0.53	0.63	0.21	0.55	0.63	0.31	0.54
	ST-FSOD [16]	0.64	0.10	0.54	0.63	0.25	0.56	0.63	0.34	0.57
	Protoalign	**0.67**	**0.15**	**0.59**	**0.68**	**0.30**	**0.60**	**0.69**	**0.38**	**0.62**
2	RAIL [65]	0.37	0.05	0.29	0.37	0.10	0.33	0.38	0.12	0.33
	TFA [8]	0.57	0.03	0.44	0.59	0.12	0.49	0.59	0.14	0.50
	iMTFA [75]	0.60	0.04	0.47	0.61	0.10	0.49	0.62	0.13	0.51
	METER [24]	0.62	0.05	0.50	0.60	0.11	0.52	0.60	0.15	0.53
	BridgeTower [25]	0.63	0.06	0.52	0.63	0.15	0.54	0.61	0.17	0.54
	DH-FSDet [76]	0.65	0.15	0.55	0.65	0.29	0.58	0.65	0.36	0.59
	G-FSDet [26]	0.63	0.16	0.54	0.63	0.37	0.57	0.64	0.42	0.56
	UMFT [20]	0.63	0.16	0.54	0.62	0.38	0.56	0.63	0.45	0.57
	ST-FSOD [16]	0.63	0.18	0.54	0.63	0.39	0.58	0.63	0.45	0.59
	Protoalign	**0.66**	**0.20**	**0.59**	**0.68**	**0.41**	**0.63**	**0.68**	**0.48**	**0.64**
3	RAIL [65]	0.35	0.03	0.30	0.36	0.07	0.31	0.36	0.08	0.31
	TFA [8]	0.59	0.04	0.48	0.61	0.06	0.50	0.62	0.05	0.51
	iMTFA [75]	0.50	0.06	0.42	0.51	0.13	0.56	0.51	0.20	0.57
	METER [24]	0.54	0.06	0.49	0.53	0.20	0.56	0.52	0.24	0.57
	BridgeTower [25]	0.57	0.09	0.50	0.58	0.23	0.56	0.57	0.26	0.57
	DH-FSDet [76]	0.66	0.10	0.54	0.67	0.20	0.58	0.68	0.23	0.59
	G-FSDet [26]	0.65	0.13	0.56	0.65	0.29	0.57	0.66	0.32	0.59
	UMFT [20]	0.65	0.12	0.55	0.65	0.30	0.58	0.64	0.34	0.61
	ST-FSOD [16]	0.66	0.14	0.56	0.66	0.31	0.59	0.66	0.33	0.60
	Protoalign	**0.70**	**0.16**	**0.63**	**0.70**	**0.33**	**0.66**	**0.72**	**0.35**	**0.68**

8) Impact of Fusion Mechanism in Integrator: We compare fusion mechanisms (i.e., element-wise fusion (EF), attention-based weighting (AW), and cross-modal fusion adapter (CFA) [77]) in the integrator on the iSAID dataset, as shown in Table IX. EF excels in efficiency with zero parameters and 2.4 GFLOPs, converging in 26 epochs, 2–3× faster than alternatives during few-shot adaptation. While it yields slightly lower novel class AP (0.16 versus 0.18 in ten shots), advanced fusion mechanism like CFA introduces significant computational costs (6.9 M parameters and 15.58 GFLOPs) for minimal performance gains. In the 50-shot setting, all methods achieve similar performance, highlighting the diminishing returns of complex fusion with more data. The substantial convergence slowdown with CFA (70 epochs) compared to EF highlights a significant tradeoff between computational complexity and training efficiency.

D. Comparison With State-of-the-Art Methods

Protoalign is compared with several state-of-the-art FSOD methods, including TFA [8], DEFRCN [75], TEMO [21], G-FSDet [26], iMTFA [29], DH-FSDET [74], FSIOD-DFP [27], GIDR [15], GE-FSOD [76], UMFT [20], BridgeTower [25], METER [24], RAIL [64], and ST-FSOD [16]. METER, RAIL, and BridgeTower have been adapted with DETR decoder for end-to-end object detection while maintaining their original image–text architectures.

1) Experiments on iSAID: Table X and Fig. 9 showcases a comprehensive evaluation of FSOD methods on iSAID across three splits, highlighting how different approaches balance catastrophic forgetting and novel class adaptation. Early detection methods like FSDetView and TFA show notable base performance instability; for instance, TFA exhibits a considerable drop in base performance (0.59–0.48 AP in split 1), indicating that they struggle to retain base knowledge while adapting to new classes. Multimodal approaches, including UMFT, METER, and BridgeTower, exhibit comparable levels of base performance degradation, suggesting that architectural complexity alone does not solve the forgetting problem. Incremental learning methods, iMTFA and G-FSDet, effectively maintain stable base performance but show moderate novel class detection, indicating a tradeoff between stability

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TABLE XI

FSOD PERFORMANCE ON THE DIOR DATASET UNDER (3-, 5-, 10-, AND 20-SHOT SETTINGS) ON FOUR DIFFERENT BASE/NOVEL SPLITS. THE BOLD DENOTES THE BEST RESULTS, AND UNDERLINE DENOTES THE SECOND BEST

Split		3 Shots			5 Shots			10 Shots			20 Shots		
	Methods	Base	Novel	All	Base	Novel	All	Base	Novel	All	Base	Novel	All
1	TFA w/cos [8]	0.70	0.11	0.55	0.70	0.11	0.56	0.70	0.15	0.57	0.71	0.18	0.58
	DEFRCN [77]	0.62	0.28	0.53	0.62	0.30	0.54	0.61	0.32	0.54	0.62	0.35	0.55
	TEMO [21]	0.61	0.23	0.52	0.64	0.31	0.56	0.65	0.35	0.58	0.65	0.40	0.59
	METER [24]	0.64	0.24	0.59	0.67	0.28	0.57	0.60	0.37	0.61	0.66	0.40	0.62
	BridgeTower [25]	0.66	0.27	0.61	0.70	0.30	0.60	0.70	0.37	0.63	0.67	0.41	0.64
	G-FSDet [26]	0.69	0.27	0.58	0.69	0.30	0.59	0.69	0.37	0.61	0.67	0.40	0.62
	FSIOD-DFP [27]	0.72	0.29	0.61	0.72	0.31	0.62	0.72	0.38	0.63	0.72	0.41	0.64
	UMFT [20]	0.72	0.30	0.61	0.73	0.32	0.63	0.73	0.38	0.64	0.72	0.40	0.63
	GE-FSOD [78]	-	0.32	-	-	0.35	-	-	0.28	-	-	0.43	-
	GIDR [15]	-	0.34	-	-	0.36	-	-	0.40	-	-	0.44	-
	ST-FSOD [16]	0.73	**0.42**	0.66	**0.78**	0.46	0.66	0.73	0.50	0.67	0.73	0.54	0.68
	Protoalign	**0.77**	0.41	**0.69**	**0.78**	**0.49**	**0.70**	**0.80**	**0.52**	**0.70**	**0.80**	**0.56**	**0.74**
2	TFA w/cos [8]	0.71	0.06	0.54	0.71	0.08	0.55	0.70	0.09	0.54	0.70	0.12	0.55
	DEFRCN [77]	0.61	0.14	0.49	0.62	0.16	0.50	0.61	0.18	0.51	0.63	0.21	0.52
	TEMO [21]	0.61	0.16	0.50	0.65	0.18	0.53	0.63	0.21	0.52	0.61	0.23	0.52
	METER [24]	0.65	0.17	0.52	0.66	0.18	0.54	0.65	0.20	0.54	0.68	0.22	0.58
	BridgeTower [25]	0.66	0.17	0.54	0.68	0.19	0.56	0.68	0.21	0.58	0.71	0.23	0.59
	G-FSDet [26]	0.69	0.14	0.55	0.69	0.16	0.57	0.68	0.20	0.56	0.68	0.22	0.57
	GE-FSOD [78]	-	0.14	-	-	0.17	-	-	0.19	-	-	0.26	-
	GIDR [15]	-	0.16	-	-	0.19	-	-	0.22	-	-	0.24	-
	FSIOD-DFP [27]	0.70	0.18	0.57	0.70	0.20	0.57	0.70	0.22	0.58	0.70	0.24	0.58
	UMFT [20]	0.70	0.19	0.58	0.71	0.23	0.59	0.71	0.22	0.60	0.72	0.26	0.61
	ST-FSOD [16]	0.73	0.18	0.59	0.73	0.21	0.60	0.72	0.27	0.61	0.73	0.33	0.63
	Protoalign	**0.77**	**0.22**	**0.63**	**0.79**	**0.24**	**0.65**	**0.79**	**0.31**	**0.68**	**0.80**	**0.35**	**0.69**
3	TFA w/cos [8]	0.72	0.08	0.56	0.71	0.10	0.56	0.72	0.10	0.57	0.73	0.18	0.59
	DEFRCN [77]	0.63	0.15	0.51	0.63	0.18	0.52	0.64	0.20	0.53	0.64	0.25	0.54
	TEMO [21]	0.66	0.16	0.53	0.66	0.23	0.55	0.68	0.25	0.57	0.68	0.31	0.58
	METER [24]	0.65	0.16	0.54	0.66	0.23	0.56	0.69	0.27	0.59	0.68	0.31	0.59
	BridgeTower [25]	0.68	0.19	0.57	0.68	0.25	0.58	0.70	0.28	0.61	0.71	0.33	0.60
	G-FSDet [26]	0.71	0.16	0.57	0.70	0.23	0.58	0.71	0.26	0.60	0.71	0.32	0.61
	FSIOD-DFP [27]	0.73	0.17	0.59	0.73	0.25	0.61	0.73	0.30	0.62	0.73	0.32	0.62
	GE-FSOD [78]	-	0.19	-	-	0.23	-	-	0.30	-	-	0.31	-
	GIDR [15]	-	0.20	-	-	0.25	-	-	0.29	-	-	0.32	-
	UMFT [20]	0.72	0.20	0.60	0.74	0.26	0.62	0.74	0.30	0.63	0.74	0.34	0.64
	ST-FSOD [16]	**0.75**	0.21	0.62	0.76	0.26	0.63	0.76	0.31	0.65	0.76	0.35	0.65
	Protoalign	**0.75**	**0.24**	**0.64**	**0.78**	**0.28**	**0.67**	**0.78**	**0.32**	**0.68**	**0.78**	**0.38**	**0.68**
4	TFA w/cos [8]	0.69	0.10	0.54	0.69	0.14	0.55	0.68	0.14	0.55	0.69	0.12	0.54
	DEFRCN [77]	0.60	0.11	0.47	0.60	0.18	0.49	0.59	0.21	0.50	0.60	0.27	0.52
	TEMO [21]	0.66	0.17	0.54	0.66	0.21	0.55	0.67	0.27	0.57	0.67	0.32	0.58
	METER [24]	0.66	0.18	0.55	0.68	0.20	0.56	0.67	0.25	0.56	0.69	0.32	0.59
	BridgeTower [25]	0.68	0.19	0.56	0.69	0.21	0.57	0.68	0.27	0.59	0.70	0.34	0.62
	G-FSDet [26]	0.69	0.17	0.56	0.68	0.21	0.56	0.68	0.26	0.58	0.67	0.32	0.58
	FSIOD-DFP [27]	0.70	0.18	0.57	0.70	0.22	0.58	0.70	0.28	0.60	0.70	0.35	0.61
	UMFT [20]	0.70	0.19	0.58	0.71	0.23	0.59	0.71	0.28	0.60	0.71	0.33	0.60
	GE-FSOD [78]	-	0.21	-	-	0.22	-	-	0.26	-	-	0.29	-
	GIDR [15]	-	0.22	-	-	0.25	-	-	0.32	-	-	0.38	-
	ST-FSOD [16]	0.73	0.20	0.60	0.74	0.25	0.61	0.74	0.33	0.64	0.74	0.38	0.65
	Protoalign	**0.76**	**0.22**	**0.64**	**0.78**	**0.27**	**0.65**	**0.80**	**0.34**	**0.67**	**0.80**	**0.41**	**0.69**

and adaptation. ST-FSOD, despite self-training, shows declining base performance while improving novel class detection (0.10–0.34 AP in split 1), highlighting the challenge of balancing pseudo-label learning with base knowledge. Across all three splits, these patterns remain consistent. In contrast, Protoalign consistently demonstrates exceptional stability and adaptation, maintaining strong base performance (0.67–0.69 AP in split 1) while significantly improving novel class detection (0.15–0.38 AP in split 1) across all splits. Notably, Protoalign achieves this strong novel class adaptation without sacrificing base performance, demonstrating that catastrophic forgetting can be effectively prevented while achieving successful feature adaptation for novel categories.

2) Experiments on DIOR: Table XI and Fig. 10 present the experimental results on the DIOR dataset, revealing critical patterns in FSOD across four splits and multiple shot settings ($3 \rightarrow 20$) shots. Similar to the results observed on the iSAID dataset, the performance patterns validate consistent method behaviors across different domains. The analysis uncovers three distinct forgetting patterns. Early methods like FRW and Meta-RCNN exhibit immediate forgetting, showing rapid base knowledge degradation even with minimal novel examples (FRW base: $0.48 \rightarrow 0.47 \rightarrow 0.46 \rightarrow 0.43$ AP in split 2). In contrast, image–text methods demonstrate delayed forgetting, maintaining base performance initially but degrading at higher shots (METER: $0.65 \rightarrow 0.66 \rightarrow 0.65 \rightarrow 0.68$ AP in split 2), suggesting that cross-modal features provide temporary protection against forgetting. Modern approaches like FSIOD-DFP achieve stable retention with consistent base performance ($0.70 \rightarrow 0.70 \rightarrow 0.70 \rightarrow 0.70$ AP) but show limited

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TABLE XII

FSOD PERFORMANCE ON THE FAIR1M-AIRPLANE DATASET UNDER (TWO-, THREE-, FIVE-, AND TEN-SHOT SETTINGS) ON ONE BASE/NOVEL SPLIT. THE BOLD DENOTES THE BEST RESULTS, AND UNDERLINE DENOTES THE SECOND BEST

Split	Methods	2 Shots			3 Shots			5 Shots			10 Shots		
		Base	Novel	All	Base	Novel	All	Base	Novel	All	Base	Novel	All
1	TFA w/cos [8]	0.50	0.29	0.45	0.52	0.31	0.47	0.54	0.31	0.48	0.55	0.30	0.48
	DEFRCN [77]	0.44	0.30	0.40	0.43	0.33	0.40	0.45	0.36	0.43	0.49	0.37	0.46
	TEMO [21]	0.50	0.31	0.45	0.50	0.33	0.45	0.51	0.37	0.48	0.52	0.41	0.49
	iMTFA [75]	0.46	0.30	0.42	0.47	0.30	0.42	0.49	0.33	0.45	0.50	0.39	0.47
	METER [24]	0.48	0.33	0.43	0.50	0.32	0.43	0.49	0.33	0.48	0.50	0.40	0.48
	BridgeTower [25]	0.50	0.31	0.46	0.49	0.33	0.47	0.48	0.35	0.48	0.50	0.42	0.48
	G-FSDet [26]	0.51	0.30	0.45	0.52	0.32	0.46	0.52	0.35	0.48	0.52	0.41	0.49
	UMFT [20]	0.56	0.33	0.48	0.55	0.38	0.49	0.56	0.42	0.52	0.55	0.45	0.53
	FSIOD-DFP [27]	0.55	0.32	0.49	0.55	0.34	**0.60**	0.55	0.40	0.51	0.55	0.43	0.52
	Protoalign	**0.60**	**0.40**	**0.54**	**0.60**	**0.43**	0.55	**0.61**	**0.47**	**0.58**	**0.62**	**0.51**	**0.61**

TABLE XIII

FSOD PERFORMANCE ON NWPU VHR-10 UNDER (3-, 5-, 10-, AND 20-SHOT SETTINGS) ON TWO DIFFERENT BASE/NOVEL SPLITS. THE BOLD DENOTES THE BEST RESULTS, AND UNDERLINE DENOTES THE SECOND BEST

Split	Methods	3 Shots			5 Shots			10 Shots			20 Shots		
		Base	Novel	All	Base	Novel	All	Base	Novel	All	Base	Novel	All
1	TFA w/cos [8]	0.89	0.09	0.65	0.90	0.09	0.65	0.90	0.09	0.66	0.71	0.11	0.66
	DEFRCN [77]	0.87	0.38	0.73	0.87	0.46	0.75	0.87	0.63	0.80	0.87	0.65	0.80
	TEMO [21]	0.86	0.58	0.78	0.85	0.67	0.80	0.87	0.75	0.83	0.89	0.76	0.85
	METER [24]	0.88	0.56	0.74	0.86	0.65	0.80	0.86	0.75	0.82	0.88	0.77	0.85
	BridgeTower [25]	0.88	0.57	0.76	0.88	0.66	0.81	0.88	0.76	0.84	0.90	0.77	0.86
	G-FSDet [26]	0.89	0.49	0.77	0.88	0.56	0.79	0.88	0.72	0.83	0.90	0.75	0.85
	ST-FSOD [16]	0.90	0.57	0.77	0.90	0.68	0.82	0.91	0.70	0.84	0.91	0.72	0.84
	FSIOD-DFP [27]	0.91	0.55	0.81	0.91	0.67	0.91	0.91	0.75	0.86	0.91	0.77	0.87
	UMFT [20]	0.91	0.58	0.79	0.91	0.68	0.83	0.92	0.79	0.88	0.92	0.79	0.88
	GE-FSOD [78]	-	0.60	-	-	0. 62	-	-	0.75	-	-	0.86	-
	Protoalign	**0.93**	**0.62**	**0.82**	**0.94**	**0.73**	**0.92**	**0.94**	**0.84**	**0.93**	**0.95**	**0.87**	**0.94**
2	TFA w/cos [8]	0.90	0.11	0.66	0.91	0.12	0.67	0.91	0.11	0.67	0.90	0.11	0.67
	DEFRCN [77]	0.88	0.39	0.74	0.88	0.46	0.75	0.88	0.54	0.78	0.88	0.57	0.79
	TEMO [21]	0.88	0.51	0.77	0.88	0.59	0.80	0.89	0.65	0.82	0.88	0.73	0.84
	METER [24]	0.89	0.48	0.75	0.87	0.60	0.81	0.90	0.67	0.82	0.87	0.74	0.85
	BridgeTower [25]	0.89	0.52	0.76	0.88	0.61	0.82	0.91	0.69	0.85	0.89	0.76	0.85
	G-FSDet [26]	0.89	0.50	0.77	0.90	0.59	0.81	0.89	0.67	0.83	0.91	0.76	0.86
	UMFT [20]	0.91	0.52	0.79	0.90	0.60	0.82	0.90	0.69	0.85	0.91	0.78	0.89
	ST-FSOD [16]	0.92	0.54	0.81	0.92	0.62	0.86	0.92	0.73	0.89	0.90	0.79	0.91
	FSIOD-DFP [27]	0.92	0.52	0.80	0.92	0.61	0.83	0.92	0.71	0.86	0.92	0.78	0.88
	Protoalign	**0.93**	**0.57**	**0.84**	**0.94**	**0.66**	**0.89**	**0.94**	**0.75**	**0.91**	**0.95**	**0.80**	**0.92**

novel class improvement. The four-split evaluation demonstrates how class distribution impacts model stability. While split 1 represents balanced scenarios, performance degradation in splits 2 and 4 reveals the crucial role of class similarity in forgetting resistance. This is particularly evident in image–text models, where semantic relationships between base and novel classes directly affect stability. Protoalign demonstrates superior and consistent performance across all splits. In split 1, it maintains strong base performance (0.77 → 0.80 AP) while achieving robust novel class adaptation (0.41 → 0.51 AP). This balanced performance extends to more challenging scenarios; even in split 4, it maintains stable base performance (0.76 → 0.80 AP) while enabling effective novel class detection (0.21 → 0.41 AP).

3) Experiments on FAIR1M-Airplane: Table XII and Fig. 11 present the evaluation results on the FAIR1M-Airplane dataset, focusing specifically on airplane classes under extremely low-shot settings (two, three, five, and 10 shots). Similar to the results observed on previous datasets, the performance patterns validate consistent method behavior across different domains. The evaluation reveals complementary strengths of different architectural approaches. While incremental learning methods like FSIOD-DFP maintain stable base performance (0.55 → 0.55 → 0.55 → 0.55 AP), their novel class adaptation is relatively limited (0.32 → 0.34 → 0.40 → 0.43 AP). In contrast, UMFT demonstrates more effective feature adaptation, achieving both strong base stability (0.56 → 0.55→ 0.56→ 0.55 AP) and superior novel class improvement (0.33 → 0.38 → 0.42 → 0.45 AP). G-FSDet shows similar patterns to FSIOD-DFP, maintaining base knowledge (0.51 → 0.52 → 0.52 → 0.52 AP) but with moderate novel class gains (0.30 → 0.32 → 0.35 → 0.41 AP). These results suggest that cross-modal learning provides advantages over incremental approaches in this single-category few-shot scenario. Protoalign continues to demonstrate superior results (base: 0.60 → 0.60 → 0.61 → 0.62 AP and novel: 0.40 → 0.43 → 0.47 → 0.51 AP). These consistent patterns across datasets, even in this specialized single-category domain with extremely few shots, provide comprehensive validation of different methodological approaches in FSOD.

4) Experiments on NWPU VHR-10: Table XIII and Fig. 12 present the evaluation results on the NWPU VHR-10 dataset under four few-shot settings (3, 5, 10, and 20 shots) across two splits. Similar to results observed on other

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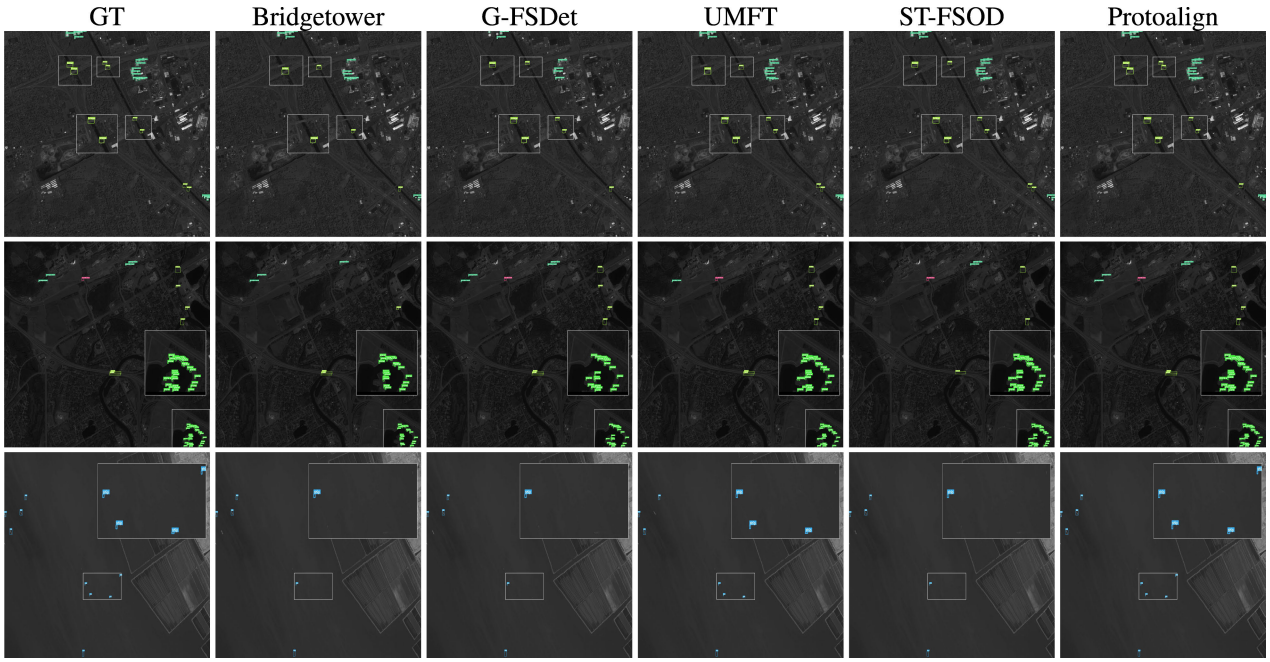


Fig. 9. FSOD visualization on iSAID under 100-shot setting.

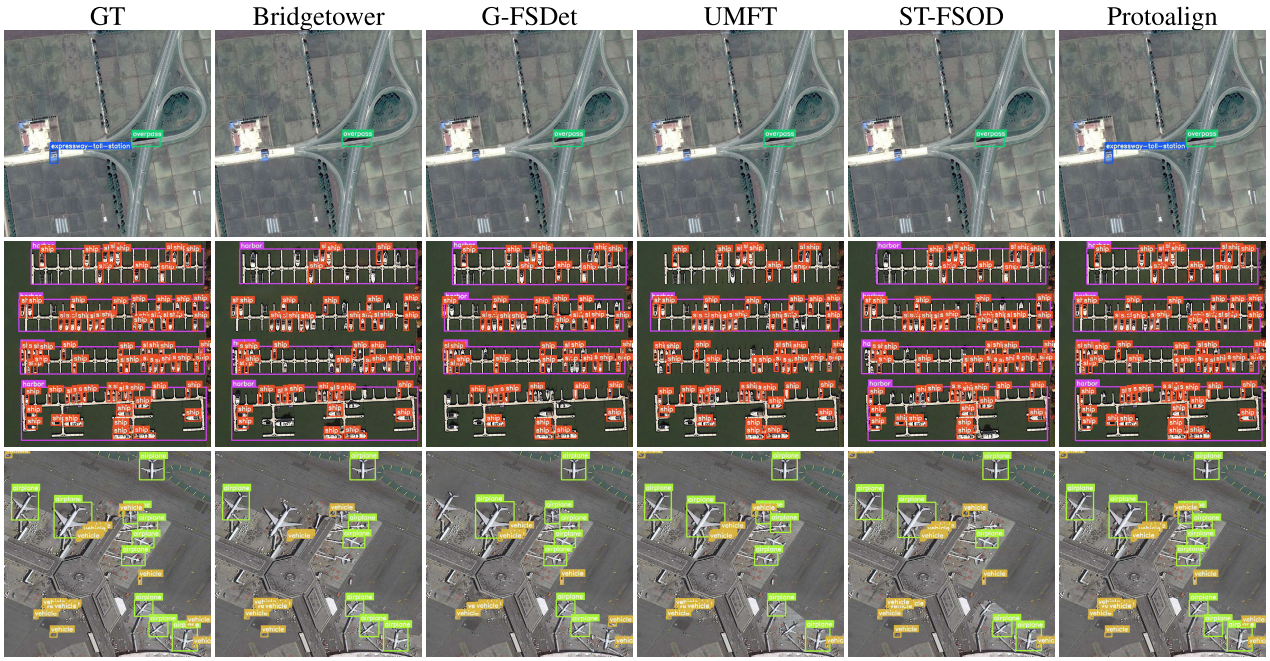


Fig. 10. FSOD visualization on the DIOR dataset under the ten-shot setting.

remote sensing datasets, the performance patterns validate consistent method behaviors across different domains. The incremental learning approaches demonstrate strong stability. G-FSDet (split 1—base: 0.89 → 0.88 → 0.88 → 0.90 AP and novel: 0.49 → 0.56 → 0.72 → 0.75 AP) and FSIOD-DFP (base: 0.91 → 0.91 → 0.91 → 0.91 AP and novel: 0.55 → 0.67 → 0.75 → 0.77 AP) both show effective resistance to catastrophic forgetting while enabling mediocre novel class adaptation, consistent with their behavior on other datasets. Image–text methods maintain their relative advantages, with UMFT (base: 0.91 → 0.91 → 0.92 → 0.92 AP and novel: 0.58 → 0.68 → 0.79 → 0.79 AP) and BridgeTower showing strong cross-modal learning capabilities. Protoalign continues to demonstrate superior results (split 1—base: 0.93 → 0.94 → 0.94 → 0.95 AP and novel: 0.62 → 0.73 → 0.84 → 0.84 AP). These consistent patterns across all four datasets, spanning natural and aerial domains with varying evaluation protocols, provide comprehensive validation of different methodological approaches in FSOD.

E. Analysis of Model Stability in FSOD

1) Impact of Cross-Modal Alignment on Catastrophic Forgetting: Catastrophic interference in image–text FSOD

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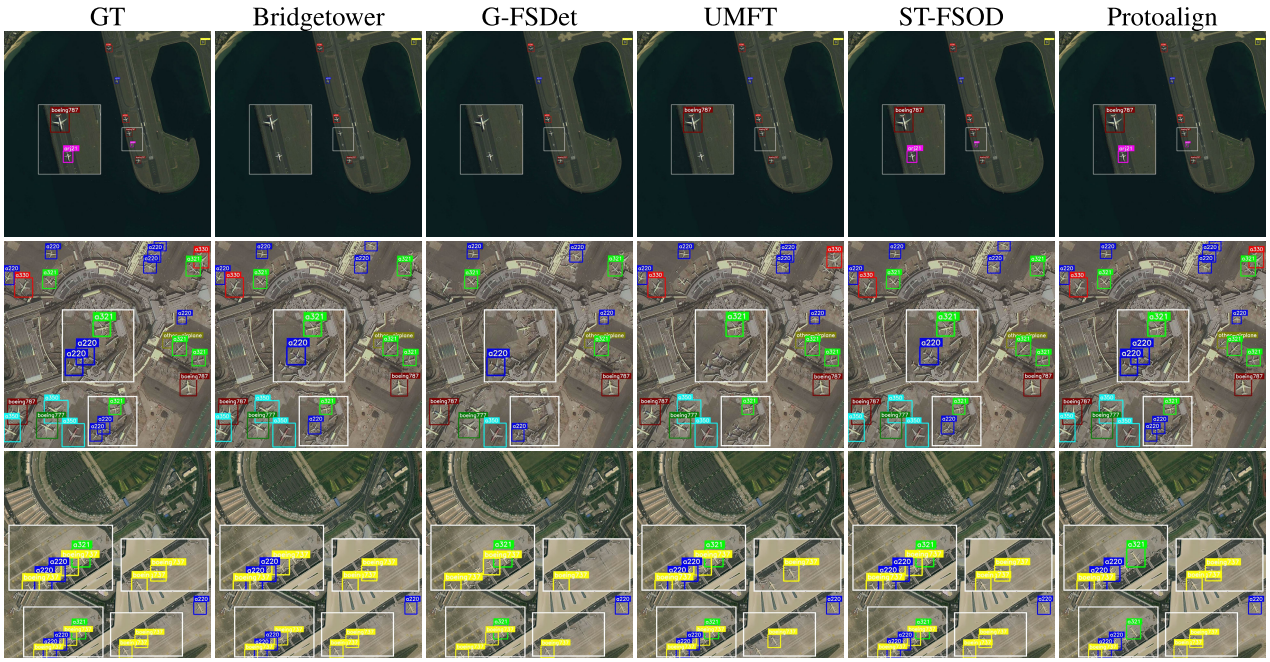


Fig. 11. FSOD visualization on the FAIR1M-Airplane dataset under the ten-shot setting.

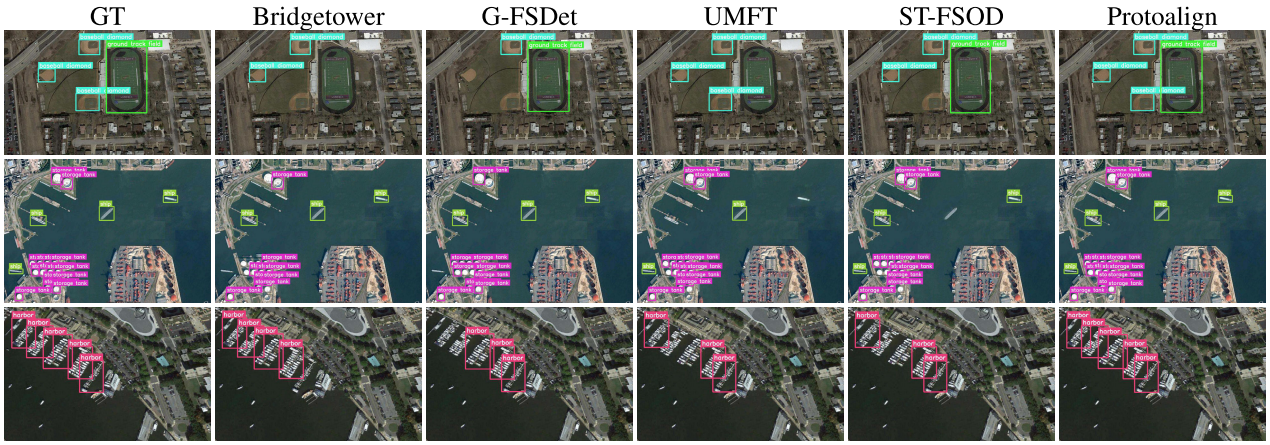


Fig. 12. FSOD visualization on the NWPU dataset under the ten-shot setting.

disrupts cross-modal feature alignments during adaptation, which we quantify using visual–textual gradient alignment (cosine similarity between cross-modal gradients pre and post-adaptation) and interference magnitude ($L2$ norm of orthogonal components between new and original cross-modal gradients). Our analysis [Fig. 13(a)] reveals that object–text complexity strongly influences interference. Object–text complexity emerges as a dominant factor, with complex objects experiencing up to 2.3× higher interference than simpler categories. Specifically, objects with intricate visual–textual relationships such as roundabouts and helicopters demonstrate severe interference (0.18 → 0.25 and 0.15 → 0.22, respectively, for Protoalign), while simpler objects like baseball diamonds exhibit substantially lower values (0.11 → 0.15). Protoalign demonstrates superior interference management, reducing cross-modal disruption by 40%–50% compared to state-of-the-art multimodal approaches like UMFT. This improvement stems from Protoalign's ability to anchor feature alignments through stable prototype representations, effectively preventing semantic drift during adaptation to novel classes.

Our temporal analysis [Fig. 13(b)] reveals distinct interference patterns across training iterations, with all object categories displaying a clear progression. During early training (1000 iterations), we observe initial interference establishment where complex objects like roundabouts already show elevated disruption (0.18) compared to simpler objects like baseball diamonds (0.11). The middle training phase (2000–3000 iterations) demonstrates progressive interference amplification, with helicopters showing the most dramatic acceleration (0.18 → 0.20) and roundabouts maintaining the highest absolute values (0.20 → 0.22). By late training (4000 iterations), interference patterns solidify with marked category-specific differences, roundabouts reach peak interference (0.25), helicopters stabilize at 0.22, and baseball diamonds maintain relatively low disruption (0.15). This confirms that cross-modal

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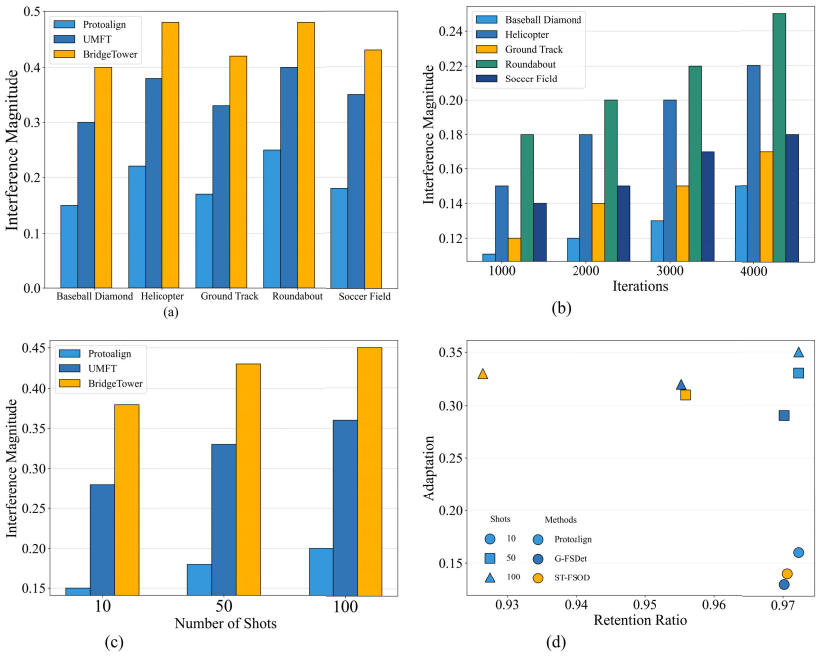


Fig. 13. Analysis of model stability in few-shot detection on iSAID split 3. (a) Peak interference. (b) Interference evolution. (c) Shot-wise interference patterns. (d) Retention versus adaptation.

interference follows a consistent trajectory across training, with early patterns becoming increasingly pronounced rather than remediated. The widening gap between simple and complex objects as training progresses (from 0.07 to 0.10 magnitude difference) suggests that longer training amplifies the interference rather than mitigating it for categories with intricate visual–textual relationships.

Our shot-wise analysis [Fig. 13(c)] demonstrates a clear nonlinear relationship between the number of shots and interference magnitude across all three architectures. As shot count increases from 10 to 100, interference magnitude grows at a decreasing rate. Protoalign shows the most controlled progression (0.15 → 0.18 → 0.20), while UMFT (0.28 → 0.33 → 0.36) and BridgeTower (0.38 → 0.43 → 0.45) exhibit steeper increases. This pattern reveals that additional training examples paradoxically amplify cross-modal disruption rather than mitigating it, with the effect being most pronounced in the 10-to-50 shot transition. Notably, Protoalign maintains consistently lower interference (55%–60% less than BridgeTower across all shot counts), demonstrating superior stability in cross-modal feature alignment. The diminishing rate of interference growth with higher shot counts suggests an approaching saturation point, indicating that architectural improvements, rather than simply adding more examples, are essential for effectively managing catastrophic forgetting in FSOD systems.

2) Knowledge Retention Versus Adaption Tradeoff: FSOD requires balancing base knowledge preservation and novel class adaptation. Our analysis of knowledge retention ratio ($P_{\text{current}}/P_{\text{initial}}$) and adaptation [Fig 13(d)] shows that Protoalign maintains a consistent retention ratio (0.97 at 100 shots) while achieving superior adaptation (0.35 AP). In contrast, G-FSDet and ST-FSOD show progressive retention degradation with increasing shot counts. At ten shots, all methods have similar retention (G-FSDet: 0.97 and ST-FSOD: 0.96), but increasing to 50 shots reveals divergent behavior, where G-FSDet's retention decreases to 0.96, while ST-FSOD's retention goes down to 0.94. This trend intensifies at 100 shots (G-FSDet: 0.95 and ST-FSOD: 0.93) with the degradation correlating inversely with novel class detection performance (G-FSDet: 0.13 → 0.29 → 0.32 AP and ST-FSOD: 0.14 → 0.31 → 0.33 AP). Protoalign achieves the most efficient tradeoff, maintaining low retention loss (below 0.01) for every 0.1 AP gain, while G-FSDet and ST-FSOD have higher retention losses for similar adaptation gains. This shows that current incremental learning and self-training approaches may prioritize feature adaptation over base knowledge. Protoalign's prototype-guided feature alignment, therefore, demonstrates superior stability, integrating novel knowledge while preserving base representations.

3) Stability Visualization in Multimodal Approaches: Fig. 14 demonstrates the effectiveness of Protoalign in mitigating catastrophic forgetting on iSAID (split 1) after few-shot adaptation. The UMFT approach demonstrates considerable limitations when adapting to novel classes with minimal training examples, generating numerous false negative detections (blue boxes) across diverse environments while still capturing true positives (red boxes). In contrast, Protoalign exhibits performance improvements by substantially reducing false negatives while preserving accurate true positive detections. This performance stability can be attributed to Protoalign's prototype-guided alignment mechanism, which effectively preserves the integrity of feature representations when learning from limited examples.

4) Failure Cases: Protoalign, despite its impressive performance in FSOD, struggles with generalization challenges as illustrated in Fig. 15. The model tends to overfit to specific

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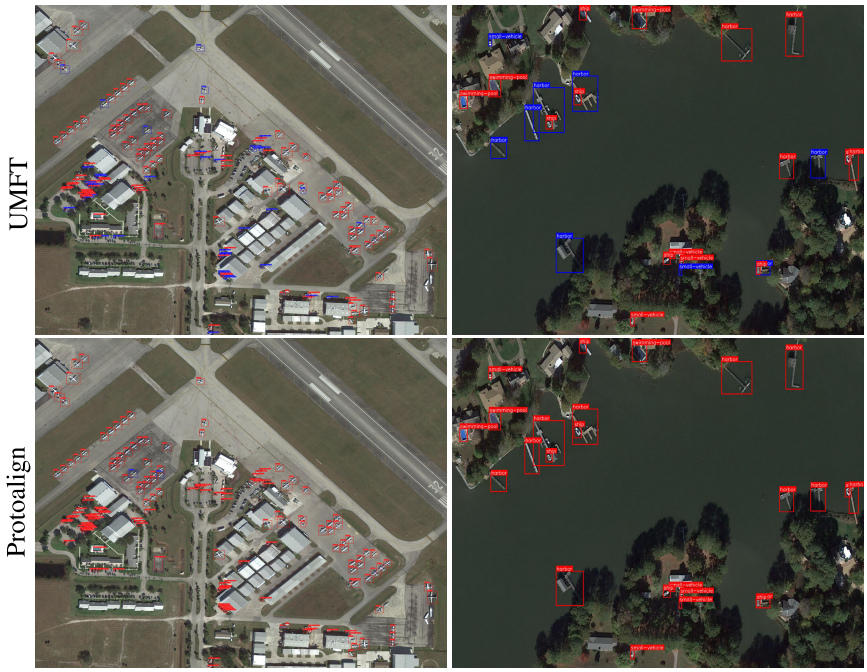


Fig. 14. Stability visualization in multimodal approaches in iSAID split 1. Red box indicates the true positive and blue box indicates the false negative.



Fig. 15. Failure cases visualization on the iSAID dataset under the 100-shot setting. White bounding boxes represent the false negative.

contextual cues from limited training examples, learning false correlations rather than robust object features. For instance, a roundabout consistently shown with similar surroundings might cause the model to rely on environmental features rather than the roundabout itself, leading to misidentification when the object appears in novel contexts. This generalization issue stems from spatial-level overfitting where the model becomes sensitive to the spatial arrangement of features and feature distribution bias where the model struggles to distinguish between characteristic and coincidental features.

V. Conclusion

In this article, we addressed a fundamental challenge of catastrophic forgetting during few-shot adaptation due to disrupted feature correspondences and misaligned visual–textual attention, ultimately compromising detection performance. We proposed Protoalign, a novel architecture that maintains robust cross-modal alignments across network depths through two mechanisms. CPG enables stable feature learning through class-specific prototypes, while MFA strengthens feature relationships through channel-level interactions. Extensive experiments on iSAID, DIOR, FAIR1M-Airplane, and NWPU VHR-10 datasets demonstrate that Protoalign achieves superior performance while significantly reducing catastrophic forgetting compared to existing methods. While Protoalign shows promising results, several directions remain for future exploration. First, incorporating an external memory module could help improve the generalization capability by storing more nuanced examples, along with their spatial and textual contexts. Moreover, integrating more advanced language models could help the model better understand complex descriptions and generalize better to varied spatial and textual contexts.

References

[1] L. Gao, D. Wang, L. Zhuang, X. Sun, M. Huang, and A. Plaza, "BS3LNet: A new blind-spot self-supervised learning network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5504218.

[2] M. Wang, L. Gao, L. Ren, X. Sun, and J. Chanussot, "Hyperspectral simultaneous anomaly detection and denoising: Insights from integrative perspective," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13966–13980, 2024.

[3] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[4] Y. Himeur, B. Rimal, A. Tiwary, and A. Amira, "Using artificial intelligence and data fusion for environmental monitoring: A review and future perspectives," *Inf. Fusion*, vols. 86–87, pp. 44–75, Oct. 2022.

[5] X. Yan, Z. Chen, A. Xu, X. Wang, X. Liang, and L. Lin, "Meta R-CNN: Towards general solver for instance-level low-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9577–9586.

[6] B. Kang, Z. Liu, X. Wang, F. Yu, J. Feng, and T. Darrell, "Few-shot object detection via feature reweighting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jun. 2019, pp. 8420–8429.

[7] L. Karlinsky et al., "RepMet: Representative-based metric learning for classification and few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5197–5206.

[8] X. Wang, T. E. Huang, T. Darrell, J. E. Gonzalez, and F. Yu, "Frustratingly simple few-shot object detection," 2020, *arXiv:2003.06957*.

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[9] H. Chen, Y. Wang, G. Wang, and Y. Qiao, "LSTD: A low-shot transfer detector for object detection," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, Apr. 2018, pp. 2836–2843.

[10] X. Li, J. Deng, and Y. Fang, "Few-shot object detection on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5601614.

[11] L. Li, X. Yao, G. Cheng, M. Xu, J. Han, and J. Han, "Solo-to-collaborative dual-attention network for one-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607811.

[12] G. Cheng et al., "Prototype-CNN for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5604610.

[13] J. Shi, Z. Jiang, and H. Zhang, "Few-shot ship classification in optical remote sensing images using nearest neighbor prototype representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 3581–3590, 2021.

[14] H. Zhang, X. Zhang, G. Meng, C. Guo, and Z. Jiang, "Few-shot multi-class ship detection in remote sensing images using attention feature map and multi-relation detector," *Remote Sens.*, vol. 14, no. 12, p. 2790, Jun. 2022.

[15] B. Yan, G. Cheng, C. Lang, Z. Huang, and J. Han, "Global-integrated and drift-rectified imprinting for few-shot remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5614611.

[16] F. Zhang, Y. Shi, Z. Xiong, and X. X. Zhu, "Few-shot object detection in remote sensing: Lifting the curse of incompletely annotated novel objects," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5603514.

[17] B. Yan, C. Lang, G. Cheng, and J. Han, "Understanding negative proposals in generic few-shot object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 5818–5829, Jul. 2024.

[18] J. Wu, C. Lang, G. Cheng, X. Xie, and J. Han, "Retentive compensation and personality filtering for few-shot remote sensing object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 7, pp. 5805–5817, Jul. 2024.

[19] C. Lang et al., "Toward open-world remote sensing imagery interpretation: Past, present, and future," *IEEE Geosci. Remote Sens. Mag.*, early access, Dec. 24, 2025, doi: 10.1109/MGRS.2024.3505602.

[20] A. Azeem, Z. Li, A. Siddique, Y. Zhang, and S. Zhou, "Unified multimodal fusion transformer for few shot object detection for remote sensing images," *Inf. Fusion*, vol. 111, Nov. 2024, Art. no. 102508.

[21] X. Lu et al., "Few-shot object detection in aerial imagery guided by text-modal knowledge," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604719.

[22] S. Zhang et al., "Text semantic fusion relation graph reasoning for few-shot object detection on remote sensing images," *Remote Sens.*, vol. 15, no. 5, p. 1187, Feb. 2023.

[23] A. Nagrani, S. Yang, A. Arnab, A. Jansen, C. Schmid, and C. Sun, "Attention bottlenecks for multimodal fusion," in *Proc. NeurIPS*, Jan. 2021, pp. 14200–14213.

[24] Z.-Y. Dou et al., "An empirical study of training end-to-end vision-and-language transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 18145–18155.

[25] X. Xu, C. Wu, S. Rosenman, V. Lal, W. Che, and N. Duan, "Bridgetower: Building bridges between encoders in vision-language representation learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 37, 2023, pp. 10637–10647.

[26] T. Zhang, X. Zhang, P. Zhu, X. Jia, X. Tang, and L. Jiao, "Generalized few-shot object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 353–364, Jan. 2023.

[27] X. Lu et al., "Few-shot incremental object detection in aerial imagery via dual-frequency prompt," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5624017.

[28] N. Dong, Y. Zhang, M. Ding, and G. H. Lee, "Incremental-DETR: Incremental few-shot object detection via self-supervised learning," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 543–551.

[29] D. A. Ganea, B. Boom, and R. Poppe, "Incremental few-shot instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1185–1194.

[30] S. W. Zamir et al., "ISAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.

[31] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[32] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[33] C. Hao, Z. Li, Y. Zhang, W. Chen, and Y. Zou, "Infrared small target detection based on adaptive size estimation by multidirectional gradient filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5007915.

[34] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[35] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[36] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[37] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6569–6578.

[38] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 213–229.

[39] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[40] H. Zhang et al., "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," in *Proc. ICLR Poster*, Feb. 2022.

[41] D. Meng et al., "Conditional DETR for fast training convergence," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3651–3660.

[42] S. Liu et al., "DAB-DETR: Dynamic anchor boxes are better queries for DETR," in *Proc. ICLR Poster*, Jan. 2022.

[43] J. Zhao et al., "OrientedFormer: An end-to-end transformer-based oriented object detector in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5640816.

[44] Y. Zeng, Y. Chen, X. Yang, Q. Li, and J. Yan, "ARS-DETR: Aspect ratio-sensitive detection transformer for aerial oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610315.

[45] Z. Hu et al., "EMO2-DETR: Efficient-matching oriented object detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5616814.

[46] Y. Liu et al., "Few-shot object detection in remote-sensing images via label-consistent classifier and gradual regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612114.

[47] Z. Yang, Y. Zhang, J. Zheng, Z. Yu, and B. Zheng, "Scale information enhancement for few-shot object detection on remote sensing images," *Remote Sens.*, vol. 15, no. 22, p. 5372, Nov. 2023.

[48] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, vol. 139, 2021, pp. 8748–8763.

[49] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024.

[50] C. Yang, Z. Li, and L. Zhang, "Bootstrapping interactive image–text alignment for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5607512.

[51] Q. Xia et al., "XGPT: Cross-modal generative pre-training for image captioning," in *Proc. 10th CCF Int. Conf. Natural Lang. Process. Chin. Comput.*, Qingdao, China. Cham, Switzerland: Springer, 2021, pp. 786–797.

[52] J. Wang et al., "GIT: A generative image-to-text transformer for vision and language," 2022, *arXiv:2205.14100*.

[53] W. Li et al., "UNIMO: Towards unified-modal understanding and generation via cross-modal contrastive learning," in *Proc. 59th Annu. Meeting Assoc. Comput. Linguistics 11th Int. Joint Conf. Natural Lang. Process.*, vol. 1, 2021, pp. 2592–2607.

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[54] Y. Ding, H. Xu, D. Wang, K. Li, and Y. Tian, "Visual selection and multistage reasoning for RSVG," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[55] Y.-C. Chen et al., "UNITER: Universal image-text representation learning," in *Proc. 16th Eur. Conf. Comput. Vis.*, Glasgow, U.K. Cham, Switzerland: Springer, Jun. 2020, pp. 104–120.

[56] G. Li, N. Duan, Y. Fang, M. Gong, and D. Jiang, "Unicoder-VL: A universal encoder for vision and language by cross-modal pre-training," in *Proc. 34th AAAI Conf. Artif. Intell.*, Jul. 2020, vol. 34, no. 7, pp. 11336–11344.

[57] X. Chen et al., "PaLI: A jointly-scaled multilingual language-image model," 2022, *arXiv:2209.06794*.

[58] L. Li, "CPSeg: Finer-grained image semantic segmentation via chain-of-thought language prompting," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 502–511.

[59] X. Xu et al., "ManagerTower: Aggregating the insights of uni-modal experts for vision-language representation learning," 2023, *arXiv:2306.00103*.

[60] M. McCloskey and N. J. Cohen, "Catastrophic interference in connectionist networks: The sequential learning problem," *Psychol. Learn. Motivat.*, vol. 24, pp. 109–165, Jan. 1989.

[61] H. Zhang et al., "Multi-stage knowledge integration of vision-language models for continual learning," 2024, *arXiv: 2411.06764*.

[62] D.-W. Zhou et al., "Learning without forgetting for vision-language models," 2023, *arXiv:2305.19270*.

[63] D. Das, D. Talon, M. Mancini, Y. Wang, and E. Ricci, "One VLM to keep it learning: Generation and balancing for data-free continual visual question answering," 2024, *arXiv:2411.02210*.

[64] Y. Xu, Y. Chen, J. Nie, Y. Wang, H. Zhuang, and M. Okumura, "Advancing cross-domain discriminability in continual learning of vision-language models," 2024, *arXiv:2406.18868*.

[65] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[66] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Amer. Chapter Assoc. Comput. Linguistics, Hum. Lang. Technol.* Minneapolis, MI, USA: Association for Computational Linguistics, vol. 1, Jun. 2019, pp. 4171–4186.

[67] Z. Jiang et al., "All tokens matter: Token labeling for training better vision transformers," in *Proc. NIPS*, vol. 34, Jul. 2021, pp. 18590–18602.

[68] R. Grainger, T. Paniagua, X. Song, N. Cuntoor, M. W. Lee, and T. Wu, "PaCa-ViT: Learning patch-to-cluster attention in vision transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 18568–18578.

[69] S. Pal et al., "OuterSPACE: An outer product based sparse matrix multiplication accelerator," in *Proc. IEEE Int. Symp. High Perform. Comput. Archit. (HPCA)*, Jun. 2018, pp. 724–736.

[70] Y. Zhan, Z. Zeng, H. Liu, X. Tan, and Y. Tian, "Mamba-SOD: Dual mamba-driven cross-modal fusion network for RGB-D salient object detection," *Neurocomputing*, vol. 631, May 2025, Art. no. 129718.

[71] K.-H. Lee, X. Chen, G. Hua, H. Hu, and X. He, "Stacked cross attention for image-text matching," in *Proc. Eur. Conf. Comp. Vis.*, 2018, pp. 201–216.

[72] X. Wang et al., "ITA: Image-text alignments for multi-modal named entity recognition," 2021, *arXiv:2112.06482*.

[73] H. Lin, X. Cheng, X. Wu, and D. Shen, "CAT: Cross attention in vision transformer," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2022, pp. 1–6.

[74] S. Wolf, J. Meier, L. Sommer, and J. Beyerer, "Double head predictor based few-shot object detection for aerial imagery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 721–731.

[75] L. Qiao, Y. Zhao, Z. Li, X. Qiu, J. Wu, and C. Zhang, "DeFRCN: Decoupled faster R-CNN for few-shot object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 8681–8690.

[76] H. Lin et al., "Generalization-enhanced few-shot object detection in remote sensing," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 6, pp. 5445–5460, Jun. 2025.

[77] A. Ghadiya, P. Kar, V. Chudasama, and P. Wasnik, "Cross-modal fusion and attention mechanism for weakly supervised video anomaly detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 1965–1974.

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